Cautious Monetary Policy*

Andrea Ferrara

Northwestern University

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Abstract

I show that when uncertainty about economic conditions is higher, the Federal Reserve responds less aggressively to changes in inflation and economic activity. I also find that, under higher uncertainty, interest rates respond less to demand shocks, while inflation and unemployment respond more. To interpret these findings, I develop a simple New Keynesian model with partial information. Higher uncertainty makes it harder for the monetary authority to determine whether a demand shock is affecting the economy, leading to a weaker policy response. As a result, a given demand shock generates larger responses of inflation and economic activity.

Keywords: Monetary Policy, Taylor Rule, Uncertainty, New Keynesian

JEL Classification Numbers: E3, E5

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1. Introduction

Monetary authorities make policy decisions under uncertainty about present and future economic conditions. Theoretical work offers conflicting predictions about how this uncertainty affects optimal monetary policy. Brainard (1967); Svensson (1999); Estrella and Mishkin (2007), among others, show that greater uncertainty can induce a more cautious policy response, while Wieland (2000); Beck and Wieland (2002); Giannoni (2002); Giannoni (2007) show that it can call for a more aggressive response. Despite these contrasting predictions, there is limited empirical evidence on how the responsiveness of monetary policy varies with the level of uncertainty.

This paper addresses the following questions: How does the level of uncertainty in the economy affect monetary policy? What are the implications for inflation and economic activity? I provide empirical evidence that the Federal Reserve adjusts interest rates more cautiously when uncertainty is higher, that is, when economic conditions are harder to predict. For a given change in inflation or economic activity, the interest rate response is smaller. I also find that, under higher uncertainty, interest rates are less sensitive to demand shocks, allowing these shocks to generate larger fluctuations in inflation and unemployment. Throughout the analysis, I treat uncertainty as a state variable that conditions monetary policy and shock propagation rather than as an exogenous shock whose direct effects are the object of interest.

I begin by estimating a Taylor rule in which inflation and the output gap are interacted with the Jurado, Ludvigson, and Ng (2015) measure of macroeconomic uncertainty. This measure summarizes forecast uncertainty about economic conditions. The interaction coefficients are negative across alternative estimation strategies, implying that the Federal Reserve's responses to inflation and the output gap decline as uncertainty rises.

To interpret this evidence, I develop a simple New Keynesian model with partial information. The economy can be in one of two regimes, one in which a demand shock affects economic conditions and another in which no demand shock is present. The monetary authority does not observe the regime and must infer it from noisy signals of inflation and the output gap. I model higher uncertainty as a proportional increase in the variances of demand shocks and signal noise. This increase raises the forecast error variances of inflation and the output gap, linking uncertainty in the model to the Jurado, Ludvigson, and Ng (2015) measure of uncertainty.

Higher uncertainty also changes what the monetary authority infers from its signals: it assigns a lower probability to the presence of a demand shock and therefore responds less aggressively to a given demand shock. The resulting optimal policy rule takes the

form of a Taylor rule in which the responses to inflation and the output gap decline as uncertainty rises, providing a theoretical counterpart to the empirical findings.

The model also predicts that demand shocks propagate more strongly when uncertainty is higher. Because the underlying demand shock is unchanged and the monetary authority responds more cautiously, inflation and the output gap must adjust by more to clear markets.

I focus on demand shocks because they move inflation and the output gap in the same direction. For given changes in both variables, the estimated Taylor rule therefore implies a smaller interest rate adjustment when uncertainty is higher, as the model predicts.

I then test the model's predictions using local projections à la Jordà (2005). I interact the business-cycle shock identified by Angeletos, Collard, and Dellas (2020) with the Jurado, Ludvigson, and Ng (2015) measure of uncertainty, allowing the responses of the policy rate, inflation, and unemployment to a given shock to vary with the level of uncertainty. I use this shock because it generates positive comovement between inflation and economic activity and induces a sizable response of the policy rate.

I find that, in response to the same expansionary demand shock, the policy rate rises less, inflation rises more, and unemployment falls more when uncertainty is higher. These results are consistent with the model's predictions and reinforce the Taylor rule evidence that the Federal Reserve responds more cautiously when uncertainty is higher.

Literature review. The literature on uncertainty and monetary policy offers contrasting theoretical predictions. Uncertainty about policy effectiveness can induce a more cautious response (Brainard, 1967; Svensson, 1999; Estrella and Mishkin, 2007), although this prediction depends on which structural parameters are uncertain (Craine, 1979; Kimura and Kurozumi, 2007). Experimentation can instead induce a more aggressive response when policy actions reveal information about policy transmission (Wieland, 2000a; Wieland, 2000b; Beck and Wieland, 2002). Robust control theory (Hansen and Sargent, 2001) can also imply either greater caution or greater aggressiveness in response to model misspecification (Giannoni, 2002; Giannoni, 2007; Tetlow and Von zur Muehlen, 2001). Finally, under partial information, optimal policy places less weight on noisier indicators (Kuttner, 1992; Rudebusch, 2001; Smets, 2002; Ehrmann and Smets, 2003; Svensson and Woodford, 2003; Aoki, 2003; Swanson, 2004; Cukierman and Lippi, 2005; Aoki, 2006; Lippi and Neri, 2007; Boehm and House, 2019).

I build on work on optimal monetary policy under partial information by allowing the monetary authority to be uncertain about whether a demand shock is present and modeling higher aggregate uncertainty as a proportional increase in demand shock and measurement error variances, leaving relative signal precision unchanged. I study how

aggregate uncertainty affects monetary policy and demand shock propagation.

The empirical analysis extends the literature on monetary policy rules (Clarida, Gali, and Gertler, 2000; Orphanides, 2001; Carvalho, Nechio, and Tristao, 2021) by allowing the Federal Reserve's responses to inflation and the output gap to vary with uncertainty. Bauer, Pflueger, and Sunderam (2024) document that professional forecasters perceive a more cautious policy rule when uncertainty rises, whereas I examine realized Federal Reserve behavior. Cieslak et al. (2023) include uncertainty as a level effect in a Taylor rule but do not examine how it changes the responses to inflation and the output gap. Unlike studies of the direct macroeconomic effects of uncertainty shocks (Bloom, 2009; Cho et al., 2021), my analysis treats uncertainty as a state variable that conditions monetary policy and shock propagation.

Outline. Section 2 presents the Taylor-rule evidence. Section 3 develops a simple New Keynesian model to interpret the findings. Section 4 examines how the effects of demand shocks vary with uncertainty. Section 5 concludes.

2. A Taylor rule that accounts for the effect of uncertainty

Since Taylor (1993), Taylor rules have served as a benchmark for describing how interest rates respond to inflation and the output gap. For instance, Carvalho, Nechio, and Tristao (2021) estimated the following specification:

$$r_t = \alpha + \beta_1 \pi_t + \gamma_1 y_t + \rho_1 r_{t-1} + \rho_2 r_{t-2} + u_t \tag{1}$$

where r_t is the federal funds rate,[1] π_t is inflation, y_t is the output gap, and u_t is the error term, which can be interpreted as a monetary policy shock.

However, equation (1), which I refer to as the "benchmark" Taylor rule, does not account for how uncertainty influences the monetary authority's interest rate decisions. To capture this effect, I include uncertainty and its interactions with inflation and the output gap:

$$\begin{aligned} r_t = \alpha + \beta_1 \pi_t + \beta_2 (\pi_t \times JLN_{t-1}) + \gamma_1 y_t + \gamma_2 (y_t \times JLN_{t-1}) \\ + \delta_1 JLN_{t-1} + \rho_1 r_{t-1} + \rho_2 r_{t-2} + u_t \end{aligned} \tag{2}$$

where JLN_{t-1} is the Jurado, Ludvigson, and Ng (2015) measure of uncertainty lagged by

[1] In the empirical analysis, I use the shadow rate of Wu and Xia (2016) when the federal funds rate is constrained by the zero lower bound to capture the stance of monetary policy, including unconventional measures such as quantitative easing.

one period.[2] I refer to this specification as the "Taylor rule considering uncertainty".

The coefficients of interest are β_1, γ_1, β_2, and γ_2. The coefficients β_1 and γ_1 capture the level effects of inflation and the output gap on the interest rate, respectively. Both are expected to be positive, as higher inflation or a higher output gap typically leads the monetary authority to raise interest rates to cool the economy. The coefficients β_2 and γ_2 capture how these responses vary with uncertainty.

The variable JLN_{t-1} enters equation (2) in levels as a control, following standard practice in specifications with interaction terms. Its coefficient, δ_1, is not central to the analysis, as the paper focuses on how uncertainty modifies the responses of monetary policy to inflation and the output gap, rather than its direct effect on the interest rate.

A challenge in estimating Taylor rules is potential endogeneity: inflation and the output gap may be correlated with the monetary policy shock. I first estimate equation (1) and equation (2) by ordinary least squares (OLS). Since monetary policy shocks account for only a small share of business-cycle fluctuations, the resulting OLS bias is expected to be limited (Carvalho, Nechio, and Tristao, 2021). I then employ two instrumental-variable strategies. Following Clarida, Gali, and Gertler (2000), I use generalized method of moments (GMM) with lagged variables as instruments, which are predetermined relative to monetary policy shocks. Following Barnichon and Mesters (2020), I also use macroeconomic shocks orthogonal to monetary policy shocks as instruments.

I use quarterly data from 1982q4 to 2018q4, described in detail in Appendix A. Table 1 reports estimates of equation (1) in columns (1), (3), and (5), and of equation (2) in columns (2), (4), and (6). Columns (1) and (2) report OLS estimates; columns (3) and (4) report GMM estimates using the same instruments as in Clarida, Gali, and Gertler (2000);[3] and columns (5) and (6) use only macroeconomic shocks as instruments. The coefficients on inflation, the output gap, uncertainty, and their interaction terms are divided by $1-\rho$, where $\rho \equiv \rho_1 + \rho_2$. This scaling allows the coefficients to be interpreted as long-run effects, following standard practice in the literature. I also standardize the JLN measure so that the interaction coefficients capture how the responses to inflation and the output gap change with a one-standard-deviation increase in uncertainty.

Columns (1), (3), and (5) report estimates of the "benchmark" Taylor rule. The coefficients on inflation and the output gap are positive and statistically significant across all three estimation methods, consistent with the literature.

Column (2) reports the OLS estimates of equation (2). Both interaction coefficients are

[2] I lag uncertainty to mitigate potential endogeneity arising from contemporaneous movements in uncertainty and the monetary policy shock.

[3] These instruments are four lags of inflation, the output gap, the policy rate, commodity price inflation, money growth, and the spread between the long-term bond rate and the three-month Treasury bill rate.

Table 1: Estimated Taylor Rule

	(1) OLS	(2) OLS	(3) GMM	(4) GMM	(5) GMM Shocks	(6) GMM Shocks
π_t^{cpi}	1.46***	1.73***	1.94***	2.10***	0.75***	1.01***
	(0.42)	(0.34)	(0.39)	(0.25)	(0.25)	(0.26)
$\pi_t^{cpi} \times JLN_{t-1}$		-1.13**		-1.12**		-0.65**
		(0.46)		(0.46)		(0.28)
y_t	1.34***	1.33***	1.01***	1.06***	1.38***	1.41***
	(0.20)	(0.18)	(0.23)	(0.15)	(0.14)	(0.09)
$y_t \times JLN_{t-1}$		-0.31**		-0.06		-0.30*
		(0.14)		(0.19)		(0.18)
JLN_{t-1}		0.87		1.69		0.01
		(1.08)		(1.39)		(1.18)
ρ	0.87***	0.85***	0.88***	0.82***	0.89***	0.87***
	(0.04)	(0.05)	(0.05)	(0.05)	(0.01)	(0.02)
N	139	139	139	139	106	106

Notes: Columns (1), (3), and (5) report estimates of equation (1), and columns (2), (4), and (6) report estimates of equation (2). Columns (1) and (2) report OLS estimates; columns (3) and (4) report GMM estimates using the same instruments as in Clarida, Gali, and Gertler (2000); and columns (5) and (6) report GMM estimates using only lagged macroeconomic shocks as instruments: the business-cycle shock of Angeletos, Collard, and Dellas (2020), the excess bond-premium shock of Gilchrist and Zakrajšek (2012), the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). These shocks enter the instrument set with lags 5 through 20. The coefficients on inflation, the output gap, uncertainty, and their interaction terms are divided by $1-\rho$, where $\rho \equiv \rho_1 + \rho_2$, and are interpreted as long-run effects. The JLN measure is standardized. Heteroskedasticity-robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

negative and statistically significant, indicating that the Federal Reserve responds less aggressively to inflation and the output gap when uncertainty is higher. The coefficients on inflation and the output gap remain close to those in the "benchmark" Taylor rule, suggesting that uncertainty modifies these responses without substantially changing the estimated level effects.

Columns (4) and (6) report the GMM estimates of equation (2). Both interaction coefficients remain negative, while the coefficients on inflation and the output gap are similar to those in columns (3) and (5), respectively. Overall, Table 1 provides evidence that the Federal Reserve responds more cautiously to inflation and the output gap when uncertainty is higher.

Robustness checks. The benchmark JLN measure uses 3-month-ahead forecasts. I consider horizons of 1 and 12 months in Table A.1 and Table A.2, respectively, and replace

JLN with the VIX in Table A.3. Table A.4 and Table A.5 report results using GDP deflator and PCE inflation, respectively. Finally, Table A.6 uses the federal funds rate throughout the sample, without substituting the shadow rate of Wu and Xia (2016) during the zero lower bound period. These robustness checks are discussed in detail in Online Appendix A.1. The interaction coefficients generally remain negative.

Furthermore, I show that the results are not driven by the rise in uncertainty during recessions. Table A.7 shows that the findings are robust to including a recession dummy and its interactions with inflation and the output gap as additional controls.

Finally, Table A.8 shows that the findings remain robust when the coefficients on the lagged interest rate are allowed to vary with uncertainty.

3. A simple New Keynesian model

In this section, I develop a simple linearized New Keynesian model to interpret the empirical findings and derive a theoretical counterpart to equation (2). The model also predicts that, under higher uncertainty, the monetary authority responds less to a given demand shock, allowing it to generate larger responses of inflation and the output gap. All proofs are provided in Appendix B.

3.1. Model Environment

Consider a dynamic New Keynesian model in which households and firms share the private-sector information set $\mathcal{I}_t^p$, while the monetary authority's information set is $\mathcal{I}_t^m$. The private sector may have the same information as the monetary authority or a finer information set. The model's structure and parameters are known to all agents.

The monetary authority solves the following problem under discretion:

$$\min_{\{i_t\}_{t=0}^{\infty}} \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t \left(\pi_t^2 + \lambda_y y_t^2\right) \middle| \mathcal{I}_0^m\right]$$

s.t.

$$\pi_t = \kappa y_t + \beta \mathbb{E}\left(\pi_{t+1} \mid \mathcal{I}_t^p\right) \tag{3}$$

$$y_t = \mathbb{E}\left(y_{t+1} \mid \mathcal{I}_t^p\right) - \frac{1}{\sigma}\left[i_t - \mathbb{E}\left(\pi_{t+1} \mid \mathcal{I}_t^p\right)\right] + a_t \tag{4}$$

where π_t denotes inflation, y_t the output gap, i_t the nominal interest rate, and a_t a demand shock. All variables are expressed as deviations from steady state. The parameter λ_y governs the relative weight on output-gap stabilization in the monetary authority's objective.

Equation (3) is the New Keynesian Phillips curve, where κ denotes its slope. Equation (4) is the Euler equation augmented with the demand shock, and σ denotes the inverse of the intertemporal elasticity of substitution. The expectation operator $\mathbb{E}(\cdot \mid \mathcal{I}_t^j) \equiv \mathbb{E}_t^j$ denotes rational expectations conditional on the information set $\mathcal{I}_t^j$, for $j \in \{m, p\}$.

Lemma 1 characterizes optimal monetary policy under partial information.

LEMMA 1. *Suppose that private agents have the same information set as the monetary authority or a finer one. Then, for any process governing a_t, the optimal interest rate i_t^* under discretion satisfies*

$$i_t^* = \sigma \mathbb{E}_t^m a_t. \tag{5}$$

Regardless of the process governing the demand shock a_t, Lemma 1 establishes certainty equivalence, a standard property of optimal monetary policy under partial information (Svensson and Woodford, 2004). Specifically, optimal policy takes the same form as with full information, except that the monetary authority responds to its estimate of the demand shock rather than to its realized value. Thus, uncertainty affects policy through the estimated demand shock, without directly changing the policy coefficients. This differs from Brainard (1967), where uncertainty about the parameter governing the effectiveness of monetary policy breaks certainty equivalence. In the present model, all structural parameters are known, and uncertainty arises because the monetary authority cannot perfectly observe the demand shock.

Lemma 1 applies to a broad class of information structures. In what follows, I specify what the monetary authority and private agents observe. The model's main innovation is to allow for two regimes. In one regime, the underlying demand shock directly affects the economy. In the other, it does not. The monetary authority must infer which regime the economy is in.

3.1.1. Serially independent demand shocks

Suppose the economy can be in one of two regimes, $s_t \in \{D, N\}$. In regime D, a demand shock affects the economy; in regime N, no demand shock is present. The monetary authority does not observe the regime and must infer whether observed movements in inflation and the output gap reflect an active demand disturbance.

The regime follows a Markov process with transition probabilities

$$\Pr(s_t = D \mid s_{t-1} = D) = p_{DD}, \qquad \Pr(s_t = N \mid s_{t-1} = N) = p_{NN}.$$

The effective demand wedge is

$$a_t \equiv \mathbf{1}\{s_t = D\} d_t, \qquad d_t \mid s_t = D, \omega_t \sim N(0, \omega_t), \qquad \omega_t > 0.$$

Here, d_t denotes the underlying demand shock, which is drawn independently each period conditional on regime D and the observed uncertainty level ω_t. Thus, the demand regime can persist even though the shock itself is serially independent.

The monetary authority observes noisy signals of the output gap and inflation:

$$\widetilde{y}_t = y_t + v_t^y, \qquad \widetilde{\pi}_t = \pi_t + v_t^\pi.$$

If the monetary authority could observe inflation and the output gap without noise, it could perfectly infer the demand shock and identify the underlying regime of the economy.

Assume that, conditional on ω_t, measurement errors are independent of the demand shock and the regime and have a common variance:

$$\begin{bmatrix} v_t^y \\ v_t^\pi \end{bmatrix} \Bigg| \omega_t \sim N(0, \chi \omega_t I_2), \qquad \chi > 0.$$

Thus, ω_t governs the overall level of uncertainty in the economy by scaling the variances of both the underlying demand shock and the measurement errors in the monetary authority's signals. The parameter χ determines the variance of measurement errors relative to that of the demand shock, with higher values implying lower signal precision. Both private agents and the monetary authority observe the realization of ω_t.

The distinctive feature of the demand process is that the monetary authority must infer whether a demand shock is present. This generates policy caution even when relative signal precision, governed by χ, remains unchanged, complementing the analysis of noisy indicators in Aoki (2003); Aoki (2006). The specification also provides a model counterpart to the measure developed by Jurado, Ludvigson, and Ng (2015) (henceforth JLN). Their measure summarizes forecast uncertainty across a broad set of macroeconomic variables. The next lemma formalizes this connection by showing how higher ω_t raises the forecast error variances of inflation and the output gap.

LEMMA 2. *Suppose uncertainty is known and constant at $\omega > 0$ over the forecast horizon. For a fixed prior probability $q_{t+1|t} \equiv \Pr(s_{t+1} = D \mid \mathcal{I}_t^m) \in (0,1)$, the one-period-ahead forecast error*

variances of inflation and the output gap are proportional to ω. Define

$$V_t^x(1) \equiv \mathbb{E}\Big[(x_{t+1} - \mathbb{E}[x_{t+1} \mid \mathcal{I}_t^m])^2 \mid \mathcal{I}_t^m\Big], \qquad x \in \{y, \pi\}.$$

Then

$$V_t^y(1) \propto \omega, \qquad V_t^\pi(1) \propto \omega,$$

Lemma 2 provides a direct link between the model's uncertainty measure ω and the forecast uncertainty measured by JLN. Their measure of macroeconomic uncertainty averages conditional forecast error standard deviations across a broad set of economic variables. Applying the same construction to inflation and the output gap gives

$$\mathcal{U}_t(1) \equiv \frac{\sqrt{V_t^y(1)} + \sqrt{V_t^\pi(1)}}{2} \propto \sqrt{\omega}.$$

Higher ω therefore corresponds to greater forecast uncertainty, consistent with the interpretation of the empirical JLN measure.

I now turn to optimal policy. Since the policy rate is observed, I can remove its contribution from the signals by defining

$$r_t^y \equiv \widetilde{y}_t + \frac{i_t}{\sigma}, \qquad r_t^\pi \equiv \widetilde{\pi}_t + \frac{\kappa i_t}{\sigma}.$$

I collect these policy-adjusted signals in $r_t \equiv (r_t^y, r_t^\pi)'$.

In the comparative statics below, I hold r_t fixed and study how uncertainty affects optimal monetary policy. The raw inflation and output gap signals also reflect the effects of monetary policy. Holding them fixed while the policy rate changes would therefore change the information about the demand shock contained in r_t. Holding r_t fixed instead lets me compare policy responses to the same signal realization across uncertainty levels. This comparison parallels the empirical exercise, in which inflation and the output gap are observed before the policy decision and are therefore taken as given when examining the policy response.

PROPOSITION 1. *The optimal interest rate is*

$$i_t^* = \sigma q(r_t, \omega_t) \mathbb{E}_t^m[d_t \mid s_t = D]. \tag{6}$$

Equivalently,

$$i_t^* = \phi_\pi(\omega_t)\widetilde{\pi}_t + \phi_y(\omega_t)\widetilde{y}_t, \tag{7}$$

where

$$\phi_y(\omega_t) = \frac{\sigma q(r_t, \omega_t)}{\chi + H - Hq(r_t, \omega_t)} > 0, \qquad \phi_\pi(\omega_t) = \kappa\phi_y(\omega_t), \qquad H = 1 + \kappa^2.$$

Here, $q(r_t, \omega_t) \equiv \Pr(s_t = D \mid \mathcal{I}_t^m)$ is the monetary authority's posterior probability that the demand regime is active. For a given policy-adjusted signal r_t and a fixed prior regime probability, this posterior probability and the policy responses to observed inflation and the output gap weakly decrease with uncertainty:

$$\frac{\partial q(r_t, \omega_t)}{\partial \omega_t} \leq 0, \qquad \frac{\partial \phi_y(\omega_t)}{\partial \omega_t} \leq 0, \qquad \frac{\partial \phi_\pi(\omega_t)}{\partial \omega_t} \leq 0.$$

Equation (6) expresses the optimal interest rate as the product of the estimated demand shock conditional on the active regime, $\mathbb{E}_t^m[d_t \mid s_t = D]$, its posterior probability, $q(r_t, \omega_t)$, and σ. For a given policy-adjusted signal, the conditional shock estimate is unchanged as uncertainty rises. Higher uncertainty instead lowers the posterior probability that the demand regime is active, leading to a weaker policy response.

Proposition 1 holds whenever the private sector has the same information set as the monetary authority or a finer one.

The representation in equation (7) expresses optimal policy in terms of the observed inflation and output gap signals. The following corollary provides a first-order approximation with respect to uncertainty, making the connection to the empirical specification in equation (2) explicit.

COROLLARY 1. *For a given policy-adjusted signal r_t, the optimal interest rate admits the following first-order approximation with respect to uncertainty ω_t:*

$$i_t^* \approx \beta_\pi \widetilde{\pi}_t + \gamma_\pi(\widetilde{\pi}_t\omega_t) + \beta_y\widetilde{y}_t + \gamma_y(\widetilde{y}_t\omega_t), \tag{8}$$

where $\beta_\pi \geq 0$, $\beta_y \geq 0$, $\gamma_\pi \leq 0$, and $\gamma_y \leq 0$.

Equation (8) has the same interaction structure as the empirical specification in equation (2), which also includes lagged interest rates and the level of uncertainty as controls. Corollary 1 therefore provides a theoretical counterpart to the empirical results in Section 2. The policy rate increases with inflation and the output gap, while both responses

decline as uncertainty rises.

Furthermore, equation (8) does not imply a first-order direct effect of uncertainty on the policy rate: when inflation and the output gap are at their steady-state values, a change in ω_t does not induce a policy adjustment. This prediction is consistent with the statistically insignificant coefficient on uncertainty reported in Table 1.

I now turn to the equilibrium responses of inflation and the output gap, characterized in Proposition 2.

PROPOSITION 2. *For a given policy-adjusted signal r_t and a fixed prior regime probability, the equilibrium output gap, y_t^*, and inflation, π_t^*, are given by*

$$y_t^* = \gamma_a(\omega_t)a_t - \gamma_v(\omega_t)(v_t^y + \kappa v_t^\pi), \qquad \pi_t^* = \kappa y_t^*,$$

where $\gamma_a(\omega_t) \geq 0$ and $\gamma_v(\omega_t) \geq 0$. Higher uncertainty weakly increases the propagation of demand shocks to inflation and the output gap and weakly reduces the propagation of measurement errors:

$$\frac{\partial \gamma_a(\omega_t)}{\partial \omega_t} \geq 0, \qquad \frac{\partial \gamma_v(\omega_t)}{\partial \omega_t} \leq 0.$$

Both comparative statics in Proposition 2 follow from the weaker policy response under higher uncertainty. The monetary authority responds less aggressively to a given demand shock, so inflation and the output gap adjust by more. Measurement errors affect equilibrium outcomes only through the interest rate response. As this response weakens, the fluctuations induced by measurement errors decline. As in Proposition 1, this result holds whenever the private sector has the same information set as the monetary authority or a finer one.

Proposition 2 also highlights a difference from the full-information benchmark. Under full information, the monetary authority perfectly offsets demand shocks, keeping inflation and the output gap at their steady-state values. This is the familiar divine-coincidence result in the standard New Keynesian model. Under partial information, the monetary authority cannot perfectly infer and offset the realized demand shock, so inflation and the output gap fluctuate even under optimal monetary policy.[4]

[4] Aoki (2003); Aoki (2006); Svensson and Woodford (2004) also show that informational frictions can prevent full stabilization.

3.1.2. Persistent demand shock

Suppose now that the demand shock follows an AR(1) process when the economy remains in regime D, so that $s_{t-1} = s_t = D$:

$$d_t = \rho d_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \omega_t \bar{\sigma}_\varepsilon^2).$$

Here, ω_t scales the innovation variance, while $\bar{\sigma}_\varepsilon^2$ determines its baseline level. If the economy transitions from regime N to regime D, a new demand episode begins with $d_t = \varepsilon_t$, independently of the shock realized in the previous episode.

This assumption simplifies the filtering problem. In regime N, the effective demand shock is $a_t = 0$, so d_t does not affect equilibrium outcomes. Because each new demand episode begins with an independent draw, the monetary authority need not carry estimates of the shock's size from one episode to the next.

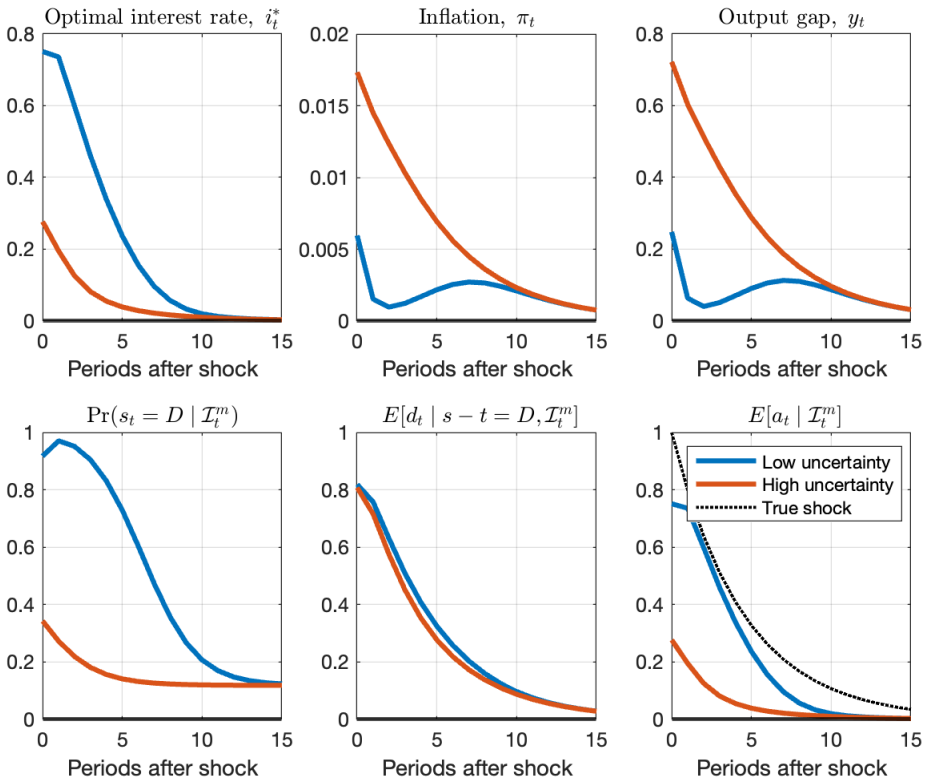
Figure 1 shows the impulse responses to a positive demand shock when regime D remains active throughout the exercise, although agents do not observe the regime. The figure compares two levels of uncertainty, ω, each held fixed over time and known to all agents.

The top left panel shows that the monetary authority raises the policy rate by less when uncertainty is higher. Inflation and the output gap rise by more, as shown in the top middle and top right panels, respectively. These responses illustrate how a weaker monetary policy response allows the demand shock to propagate more strongly through the economy.

The bottom panels of Figure 1 illustrate the mechanism underlying these responses. The bottom left panel shows that the monetary authority's posterior probability of the active demand regime, $\Pr(s_t = D \mid \mathcal{I}_t^m)$, is lower under high uncertainty. The bottom middle panel shows that the estimated demand shock conditional on the active regime, $\mathbb{E}[d_t \mid s_t = D, \mathcal{I}_t^m]$, is very similar across the two uncertainty levels. This similarity reflects the proportional scaling of the demand innovation and measurement error variances, which leaves relative signal precision unchanged. Consequently, the estimated effective demand shock, $\mathbb{E}[a_t \mid \mathcal{I}_t^m]$, shown in the bottom right panel, is smaller under higher uncertainty because the monetary authority assigns a lower probability to the active demand regime.

The exercise in Figure 1 assumes that the private sector and the monetary authority share the same information set. In Online Appendix B.5, I show that the main findings also hold when the private sector has full information and the monetary authority observes a noisy signal of the demand shock directly. This signal keeps the information

Figure 1: Impulse responses to a persistent demand shock under low and high uncertainty



Notes: The blue line denotes low uncertainty and the red line high uncertainty. The top panels report the responses of the policy rate, inflation, and the output gap. The bottom panels report the monetary authority's posterior probability of the active demand regime, $\Pr(s_t = D \mid \mathcal{I}_t^m)$, the perceived demand shock conditional on the active regime, $\mathbb{E}[d_t \mid s_t = D, \mathcal{I}_t^m]$, and the estimated effective demand shock, $\mathbb{E}[a_t \mid \mathcal{I}_t^m]$. Higher uncertainty lowers the posterior probability of the active demand regime and, consequently, the estimated effective demand shock. The monetary authority therefore responds less to the same realized demand shock, resulting in larger responses of inflation and the output gap.

available to the monetary authority independent of its contemporaneous policy choice. The appendix also describes the filtering and solution methods.

4. Local projections

Motivated by the model in Section 3, I now test whether the propagation of a given demand shock varies with the level of uncertainty. For each horizon j, I estimate the following local projection:

$$\Delta x_{t+j,t-1} = \alpha_j + \beta_{1,j}^x Shock_t + \beta_{2,j}^x (JLN_{t-1} \times Shock_t) + \beta_{3,j}^x JLN_{t-1} + \delta_j^x W_{t-1}^x + u_{t+j}, \quad (9)$$

where x_t denotes the outcome variable and $\Delta x_{t+j,t-1} \equiv x_{t+j} - x_{t-1}$. The variable $Shock_t$ is the standardized business-cycle unemployment shock identified by Angeletos, Collard, and Dellas (2020), and JLN_{t-1} is the JLN measure of uncertainty lagged by one period. The control vector W^x_{t-1} includes two lags of the dependent variable and one lag each of the shock, inflation, GDP, investment, unemployment, and the policy rate. I estimate equation (9) separately for the policy rate, unemployment, and inflation using data from 1982Q4 to 2019Q4. Appendix A describes the data.

I estimate equation (9) by OLS and use heteroskedasticity-robust standard errors. The coefficient of interest, $\beta^x_{2,j}$, captures how the response to the demand shock at horizon j varies with uncertainty. Although JLN_{t-1} need not be exogenous, Bun and Harrison (2019) show that OLS can consistently estimate the interaction coefficient when one interacted variable is endogenous and the other exogenous. Here, $Shock_t$ is the exogenous variable. For interpretability, I normalize $Shock_t$ so that a positive realization generates an increase in the federal funds rate.

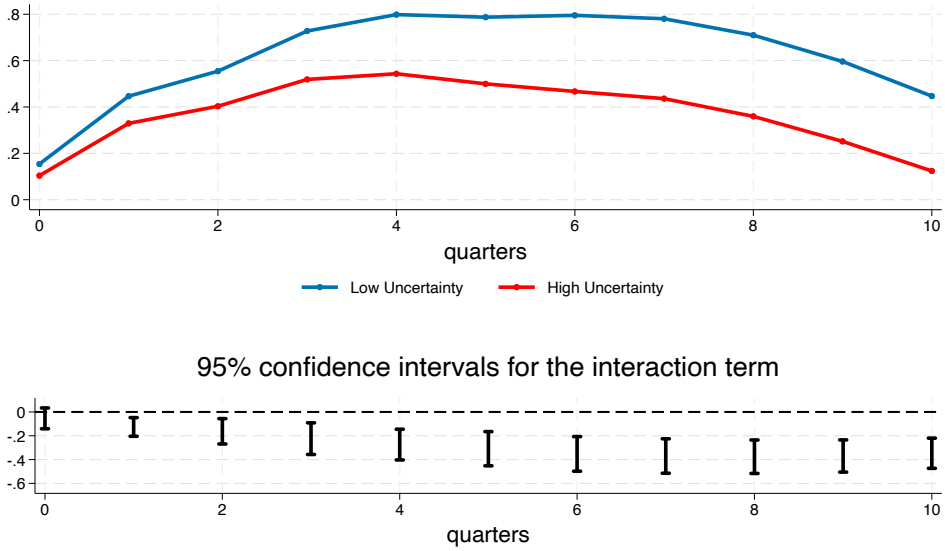
Throughout this exercise, JLN_{t-1} is a conditioning variable rather than an identified uncertainty shock. Accordingly, I do not give a causal interpretation to the coefficient $\beta^x_{3,j}$. The interaction coefficient $\beta^x_{2,j}$ characterizes the average state dependence of the response over the sample and need not describe the response in every particular episode.

I use the business-cycle unemployment shock identified by Angeletos, Collard, and Dellas (2020) as my benchmark shock for three reasons. First, it is constructed as a linear combination of fundamental shocks that maximizes the share of unemployment variance explained at business-cycle frequencies and therefore captures a quantitatively important source of cyclical variation. Second, the shock induces a sizable response of the federal funds rate, providing sufficient variation to study how this response varies with uncertainty. Third, the shock generates a boom in which both the output gap and inflation increase. This comovement calls for a higher policy rate according to the Taylor-rule logic of Section 2 and connects the local-projection exercise to the demand-shock mechanism in Section 3.

Figure 2 reports the results of estimating equation (9) using the policy rate as the dependent variable. The top panel plots the estimated response $\hat{\beta}^i_{1,j}\overline{Shock} + \hat{\beta}^i_{2,j}(\widetilde{JLN} \times \overline{Shock})$ for horizons $j = 0, \ldots, 10$. Here, $\overline{Shock}$ denotes one standard deviation of the shock and $\widetilde{JLN}$ is set to the 25th or 75th percentile of the JLN distribution, corresponding to low and high uncertainty, respectively.

The policy rate response is consistently smaller under high uncertainty: for a given demand shock, the Federal Reserve raises the policy rate by less, consistent with the model's prediction. The bottom panel shows that the 95% confidence intervals for $\hat{\beta}^i_{2,j}$

Figure 2: Local projections of the federal funds rate



Notes: The figure reports estimates of equation (9) using the federal funds rate as the dependent variable. The top panel plots $\hat{\beta}^i_{1,j}\overline{Shock} + \hat{\beta}^i_{2,j}(\widetilde{JLN} \times \overline{Shock})$ for horizons $j = 0, \ldots, 10$. $\overline{Shock}$ denotes one standard deviation of the business-cycle unemployment shock, and $\widetilde{JLN}$ is set to the 25th percentile of the JLN distribution for the low-uncertainty response and to the 75th percentile for the high-uncertainty response. The bottom panel reports the 95% confidence intervals for $\hat{\beta}^i_{2,j}$.

exclude zero at nearly all horizons, indicating statistically significant differences between the responses under low and high uncertainty.

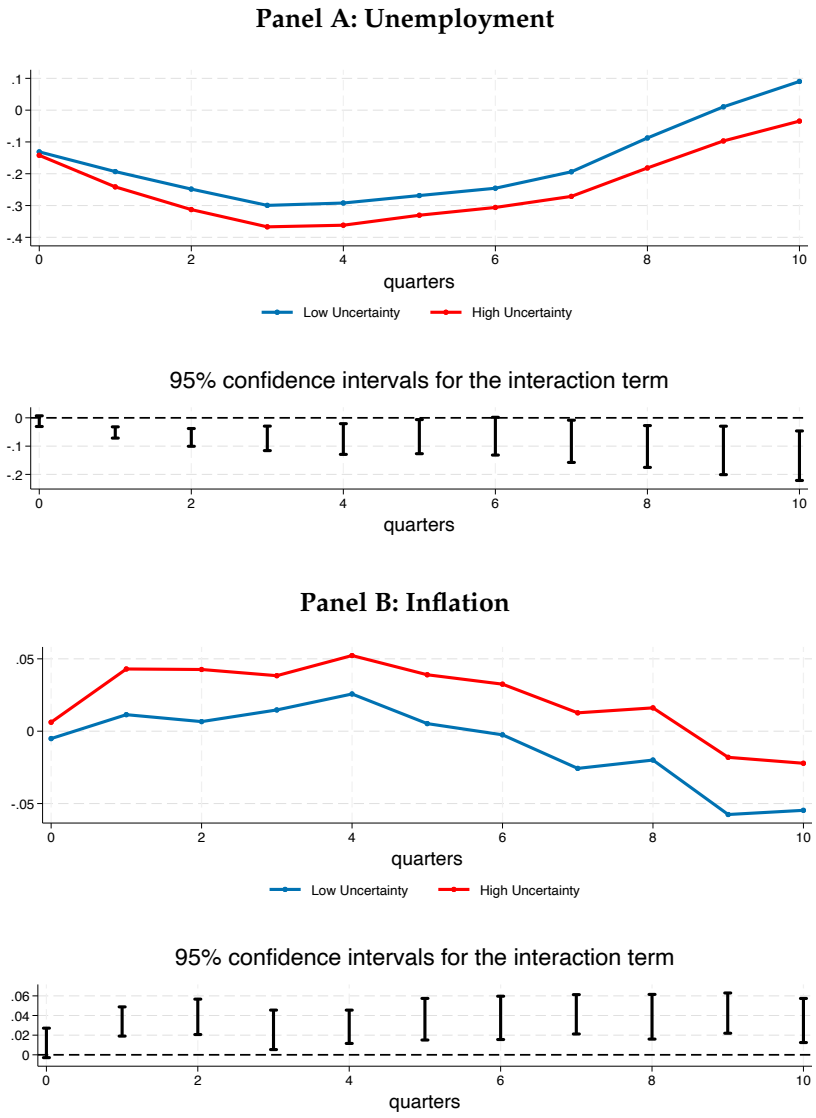
Figure 3 reports estimates of equation (9) using unemployment and inflation as the dependent variables in Panels A and B, respectively. Each panel is constructed analogously to Figure 2: the upper plot shows responses under low and high uncertainty, while the lower plot reports the 95% confidence intervals for the corresponding interaction coefficient.

Panel A shows that the unemployment response is lower under high uncertainty throughout the response horizon. Thus, the same demand shock generates a stronger expansion in economic activity under high uncertainty.

Panel B shows that the inflation response is higher under high uncertainty. Under low uncertainty, the inflation response is small and occasionally negative, reflecting the shock's modest overall effect on inflation. Overall, higher uncertainty strengthens both the expansionary and inflationary effects of the shock.

The lower plots show that the 95% confidence intervals for $\hat{\beta}^u_{2,j}$ and $\hat{\beta}^\pi_{2,j}$ exclude zero at nearly all horizons, indicating that the responses under low and high uncertainty differ

Figure 3: Local projections of unemployment and inflation



Notes: Panels A and B report estimates of equation (9) using unemployment and inflation as the dependent variable, respectively. Within each panel, the upper plot shows $\hat{\beta}^x_{1,j}\overline{Shock} + \hat{\beta}^x_{2,j}(\widetilde{JLN} \times \overline{Shock})$ for horizons $j = 0, \ldots, 10$, where $x = u$ in Panel A and $x = \pi$ in Panel B. $\overline{Shock}$ denotes one standard deviation of the business-cycle unemployment shock, and $\widetilde{JLN}$ is set to the 25th percentile of the JLN distribution for the low-uncertainty response and to the 75th percentile for the high-uncertainty response. Within each panel, the lower plot reports the 95% confidence intervals for $\hat{\beta}^x_{2,j}$.

significantly for each variable. These findings are consistent with the model's prediction that, under higher uncertainty, the policy rate responds less to a given demand shock, while inflation and economic activity respond more.

Robustness checks. I first re-estimate equation (9) without the controls W_{t-1}^{x}. Figure A.1 and Figure A.2 show that the results remain qualitatively similar to the benchmark estimates.

I then replace the JLN measure of uncertainty with the VIX. Figure A.3 and Figure A.4 show that the policy-rate and unemployment responses remain qualitatively similar to the benchmark estimates. For inflation, the difference across uncertainty states is less pronounced, although the response generally remains larger under high uncertainty.

Finally, I replace the business-cycle unemployment shock from Angeletos, Collard, and Dellas (2020) with the excess bond premium shock from Gilchrist and Zakrajšek (2012). Figure A.5 and Figure A.6 show qualitatively similar results, although the difference in unemployment responses across uncertainty states is less pronounced at longer horizons.

5. Conclusions

I document that the Federal Reserve responds less aggressively to inflation and the output gap when uncertainty is higher. Under higher uncertainty, the policy rate also responds less to a given demand shock, while inflation and economic activity respond more. A simple New Keynesian model with partial information provides an interpretation of these findings. The model links higher uncertainty to larger forecast error variances of inflation and the output gap, providing a counterpart to the empirical JLN measure. Higher uncertainty lowers the probability the monetary authority assigns to the presence of a demand shock, leading to a weaker optimal policy response and stronger shock propagation.

The model's mechanism complements the classic channel in Brainard (1967), through which uncertainty about policy effectiveness can induce caution. An increase in the volatility of fundamental macroeconomic shocks alone may also shape how monetary policy varies with uncertainty. Assessing the relative importance of these three mechanisms would require separate measures of uncertainty. Developing such measures is a natural direction for future research.

Appendix

A. Data

This section describes the data used. The data downloaded from the historical archive of FRED labeled as real time data are the following:

1. The interest rate is the federal funds rate observed in the last day of each quarter. For the lower bound period, 2009q1 - 2015q4, the federal funds rate is replaced by the shadow rate of Wu and Xia (2016).

2. CPI inflation is constructed as the annualized growth of the CPI index for each quarter.

3. The output gap is constructed as realized output minus potential output estimated by the CBO.

I complement this dataset with the following variables:

1. JLN measure is the uncertainty measure of Jurado, Ludvigson, and Ng (2015): it is the last monthly observation in each quarter.

2. The instruments for the GMM are taken from Carvalho, Nechio, and Tristao (2021).

3. VIX is the CBOE volatility index and it is constructed as the average of the last month in each quarter.

4. I extended the sample period of Angeletos, Collard, and Dellas (2020) for constructing the business cycle shock. See their online appendix and the replication package for a detailed explanation of the data used. These same data are also used as controls. Furthermore, unemployment and inflation constructed this way are used as dependent variables in Section 4.

5. The excess bond premium shock is taken from Gilchrist and Zakrajšek (2012).

6. The news oil shock is taken from Känzig (2021).

7. The government-spending shock is taken from Ramey (2016).

B. Proofs

This appendix provides proofs of Lemma 1, Proposition 1, Corollary 1, Proposition 2, and Lemma 2. More detailed derivations appear in Online Appendix B.

B.1. Proof of Lemma 1

Under discretion, optimality requires $\mathbb{E}_t^m[\kappa\pi_t + \lambda_y y_t] = 0$. Since $\mathcal{I}_t^m \subseteq \mathcal{I}_t^p$, iterated expectations applied to equation (3), together with this condition, give

$$\mathbb{E}_t^m \pi_t = \frac{\beta\lambda_y}{\kappa^2 + \lambda_y} \mathbb{E}_t^m \pi_{t+1}.$$

The coefficient is below one, so forward iteration in a bounded equilibrium implies $\mathbb{E}_t^m \pi_t = 0$ at every date. The Phillips curve then gives $\mathbb{E}_t^m y_t = 0$. Iterated expectations therefore imply $\mathbb{E}_t^m \pi_{t+1} = \mathbb{E}_t^m y_{t+1} = 0$. Taking the monetary authority's conditional expectation of equation (4) yields $i_t^* = \sigma \mathbb{E}_t^m a_t$.

B.2. Proof of Proposition 1

Write $q_t = q(r_t, \omega_t)$, let $p_t = \Pr(s_t = D \mid \mathcal{I}_{t-1}^m) \in (0,1)$, and define $S_t = r_t^y + \kappa r_t^\pi$ and $M = \chi + H$. Under serial independence, the policy-adjusted signals satisfy $r_t^y = a_t + v_t^y$ and $r_t^\pi = \kappa a_t + v_t^\pi$. Gaussian updating gives

$$\log \frac{q_t}{1-q_t} = \log \frac{p_t}{1-p_t} - \frac{1}{2}\log \frac{M}{\chi} + \frac{S_t^2}{2\omega_t \chi M},$$
$$\mathbb{E}_t^m[d_t \mid s_t = D] = \frac{S_t}{M}.$$

By Lemma 1, $i_t^* = \sigma q_t S_t / M$, establishing equation (6). Substituting $S_t = \widetilde{y}_t + \kappa\widetilde{\pi}_t + H i_t^*/\sigma$ and rearranging yields

$$i_t^* = \frac{\sigma q_t}{M - H q_t}(\widetilde{y}_t + \kappa\widetilde{\pi}_t),$$

which establishes equation (7). Both policy coefficients are positive since $q_t > 0$, $\kappa > 0$, and $M - Hq_t \geq \chi > 0$.

Holding r_t and p_t fixed,

$$\frac{\partial q_t}{\partial \omega_t} = -\frac{q_t(1-q_t)S_t^2}{2\omega_t^2 \chi M} \leq 0, \qquad \frac{\partial \phi_y}{\partial \omega_t} = \frac{\sigma M}{(M - Hq_t)^2}\frac{\partial q_t}{\partial \omega_t} \leq 0.$$

Since $\phi_\pi = \kappa\phi_y$, the same sign holds for $\partial\phi_\pi/\partial\omega_t$. The inequalities are strict whenever $S_t \neq 0$.

B.3. Proof of Corollary 1

Fix r_t and an interior regime prior. For $j \in \{\pi, y\}$, a first-order Taylor expansion around $\bar{\omega} > 0$ gives

$$\phi_j(\omega_t) \approx \underbrace{\phi_j(\bar{\omega}) - \bar{\omega}\phi_j'(\bar{\omega})}_{\beta_j} + \underbrace{\phi_j'(\bar{\omega})}_{\gamma_j}\omega_t.$$

Substitution into equation (7) yields equation (8). By Proposition 1, $\beta_j > 0$ and $\gamma_j \leq 0$, with $\gamma_j < 0$ whenever $r_t^y + \kappa r_t^\pi \neq 0$.

B.4. Proof of Proposition 2

Fix r_t and an interior regime prior, and write $q_t = q(r_t, \omega_t)$ and $M = \chi + H$. Since $S_t = r_t^y + \kappa r_t^\pi = Ha_t + v_t^y + \kappa v_t^\pi$, substituting $i_t^* = \sigma q_t S_t / M$ into $y_t = a_t - i_t/\sigma$ gives

$$y_t^* = \underbrace{\left(1 - \frac{Hq_t}{M}\right)}_{\gamma_a(\omega_t)} a_t - \underbrace{\frac{q_t}{M}}_{\gamma_v(\omega_t)} (v_t^y + \kappa v_t^\pi), \qquad \pi_t^* = \kappa y_t^*.$$

Both coefficients are positive because $0 < q_t < 1$ and $M > H$. By Proposition 1,

$$\gamma_a'(\omega_t) = -\frac{H}{M}\frac{\partial q_t}{\partial \omega_t} \geq 0, \qquad \gamma_v'(\omega_t) = \frac{1}{M}\frac{\partial q_t}{\partial \omega_t} \leq 0.$$

Both inequalities are strict whenever $S_t \neq 0$.

B.5. Proof of Lemma 2

As established in the proof of Lemma 1,

$$\mathbb{E}_t^m y_{t+1} = \mathbb{E}_t^m \pi_{t+1} = 0.$$

Fix the prior $q_{t+1|t}$ and all other model parameters. For the same prior, the Bayesian updating formula in the proof of Proposition 1 gives

$$q(r_{t+1}, \omega) = q\left(\frac{r_{t+1}}{\sqrt{\omega}}, 1\right).$$

Thus, the posterior probability depends only on the normalized signals and the fixed prior. Since both the demand shock and measurement error variances scale with ω, the conditional joint distribution of

$$\left(\frac{a_{t+1}}{\sqrt{\omega}}, \frac{v^y_{t+1}}{\sqrt{\omega}}, \frac{v^\pi_{t+1}}{\sqrt{\omega}}, q(r_{t+1}, \omega)\right)$$

given $\mathcal{I}^m_t$ is independent of ω.

Writing $q = q(r_{t+1}, \omega)$ and $H = 1 + \kappa^2$, the equilibrium expression derived in Proposition 2 implies

$$\frac{y_{t+1}}{\sqrt{\omega}} = \left(1 - \frac{Hq}{\chi + H}\right)\frac{a_{t+1}}{\sqrt{\omega}} - \frac{q}{\chi + H}\frac{v^y_{t+1} + \kappa v^\pi_{t+1}}{\sqrt{\omega}}.$$

Its conditional distribution is therefore independent of ω. Consequently,

$$C \equiv \mathbb{E}^m_t\left[\left(\frac{y_{t+1}}{\sqrt{\omega}}\right)^2\right]$$

is independent of ω. Moreover, $C > 0$ because the active regime has positive prior probability.

Using the zero conditional means and $\pi_{t+1} = \kappa y_{t+1}$,

$$V^y_t(1) = \omega C, \qquad V^\pi_t(1) = \kappa^2 \omega C.$$

This establishes both proportionalities.

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ONLINE APPENDIX

Cautious Monetary Policy

Andrea Ferrara

Northwestern University

November 9, 2026

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A. Robustness checks

This section reports the results of the robustness checks discussed in the paper.

A.1. Robustness of the Taylor rule

This section reports the results of the robustness checks discussed in Section 2. Except for Table A.8, all tables follow the structure of Table 1: columns (1), (3), and (5) report estimates of the benchmark Taylor rule, while columns (2), (4), and (6) report estimates of the Taylor rule considering uncertainty. Columns (1) and (2) report OLS estimates; columns (3) and (4) report GMM estimates using the same instruments as in Clarida, Gali, and Gertler (2000); and columns (5) and (6) report GMM estimates using only lagged macroeconomic shocks as instruments. The uncertainty measures are standardized, and the coefficients on inflation, the output gap, uncertainty, and their interaction terms are scaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, to report long-run effects.

The benchmark JLN measure of uncertainty is based on 3-month-ahead forecasts. To assess the robustness of the findings, I also consider forecast horizons of 1 and 12 months, with results reported in Table A.1 and Table A.2, respectively. The findings remain robust: both interaction coefficients are negative, and the coefficient on the interaction with inflation is always statistically significant.

The VIX can also serve as a measure of uncertainty and is often used to identify uncertainty shocks.[5] However, I do not use it as the benchmark measure because it primarily reflects stock market volatility rather than broader economic uncertainty.

Nevertheless, the VIX provides an alternative measure for assessing how uncertainty affects the Federal Reserve's Taylor rule. Table A.3 reports results using the VIX in place of the JLN measure. The sample begins in 1990q1 due to data availability. The findings remain robust: both interaction coefficients are negative and statistically significant.

Studies estimating Taylor rules use different measures of inflation. For example, Clarida, Gali, and Gertler (2000) used GDP deflator inflation, while Carvalho, Nechio, and Tristao (2021) used inflation measured by the personal consumption expenditures (PCE) price index or the consumer price index (CPI). Table A.4 and Table A.5 report results using GDP deflator inflation and PCE inflation, respectively. In both cases, the interaction coefficients remain negative, supporting the robustness of the findings.

I use CPI inflation as the benchmark measure because estimating the "benchmark" Taylor rule, equation (1), with this measure yields a larger positive and statistically sig-

[5] See, for example, Bloom (2009).

nificant coefficient on inflation.

In the benchmark specification, I replace the federal funds rate with the shadow rate of Wu and Xia (2016) from 2009q1 to 2015q4. This substitution is standard in the literature because the federal funds rate was constrained by the zero lower bound during this period. The shadow rate captures the effects of unconventional monetary policy, such as quantitative easing, providing a more accurate measure of the monetary policy stance.

Table A.6 reports results using the federal funds rate over the entire sample period. The interaction coefficients remain negative, supporting the robustness of the findings. The coefficients on inflation and the output gap are smaller in magnitude, consistent with the lower variation in the federal funds rate relative to the shadow rate.

I also examine whether the findings are driven by the rise in uncertainty during recessions. To this end, I include a lagged recession dummy and its interactions with inflation and the output gap as additional controls. Table A.7 reports the results. Both interaction coefficients with uncertainty remain negative, and the interaction with inflation is statistically significant across all three estimation methods. These results indicate that the more cautious monetary policy response under higher uncertainty is not simply capturing differences in the Federal Reserve's responses during recessions.

Finally, I allow interest-rate smoothing to vary with uncertainty by adding interactions between each of the two lagged interest rates and JLN_{t-1} to equation (2):

$$\begin{aligned} r_t = \alpha + \beta_1 \pi_t + \beta_2 (\pi_t \times JLN_{t-1}) + \gamma_1 y_t + \gamma_2 (y_t \times JLN_{t-1}) + \delta_1 JLN_{t-1} \\ + \rho_1 r_{t-1} + \rho_2 r_{t-2} + a_1 (r_{t-1} \times JLN_{t-1}) + a_2 (r_{t-2} \times JLN_{t-1}) + u_t. \end{aligned} \tag{10}$$

The coefficients a_1 and a_2 capture how interest-rate smoothing varies with uncertainty. Holding uncertainty fixed at JLN, total persistence is $\rho(JLN) = \rho_1 + \rho_2 + (a_1 + a_2)JLN$. Table A.8 reports the conditional long-run responses to inflation and the output gap, $(\beta_1 + \beta_2 JLN)/[1 - \rho(JLN)]$ and $(\gamma_1 + \gamma_2 JLN)/[1 - \rho(JLN)]$, respectively, evaluated at $JLN = -1, 0, 1$. Under the benchmark standardization, these values correspond to one standard deviation below the mean, the mean, and one standard deviation above the mean. The rescaling therefore accounts for the effect of uncertainty on both the contemporaneous response and interest-rate persistence.

The three columns report OLS and the two GMM specifications, with standard errors computed using the delta method. Both estimated responses decline as uncertainty increases under all three estimation methods, although the inflation responses at low uncertainty are imprecisely estimated.

Table A.1: Taylor rule using the 1-month JLN measure

	(1) OLS	(2) OLS	(3) GMM	(4) GMM	(5) GMM Shocks	(6) GMM Shocks
π_t^{cpi}	1.46***	1.70***	1.94***	2.08***	0.75***	1.00***
	(0.42)	(0.34)	(0.39)	(0.25)	(0.25)	(0.27)
$\pi_t^{cpi} \times JLN_{t-1}^{1m}$		-1.12**		-1.12**		-0.62**
		(0.46)		(0.46)		(0.29)
y_t	1.34***	1.32***	1.01***	1.06***	1.38***	1.39***
	(0.20)	(0.18)	(0.23)	(0.15)	(0.14)	(0.09)
$y_t \times JLN_{t-1}^{1m}$		-0.31**		-0.07		-0.30
		(0.14)		(0.19)		(0.19)
JLN_{t-1}^{1m}		0.83		1.61		-0.11
		(1.08)		(1.36)		(1.23)
ρ	0.87***	0.85***	0.88***	0.82***	0.89***	0.87***
	(0.04)	(0.05)	(0.05)	(0.05)	(0.01)	(0.02)
N	139	139	139	139	106	106

Notes: This table uses the JLN measure based on 1-month-ahead forecasts. Columns (1), (3), and (5) report estimates of equation (1), and columns (2), (4), and (6) report estimates of equation (2). Columns (1) and (2) report OLS estimates; columns (3) and (4) report GMM estimates using the same instruments as in Clarida, Gali, and Gertler (2000); and columns (5) and (6) report GMM estimates using only lagged macroeconomic shocks as instruments: the business-cycle shock of Angeletos, Collard, and Dellas (2020), the excess bond-premium shock of Gilchrist and Zakrajšek (2012), the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). These shocks enter the instrument set with lags 5 through 20. The coefficients on inflation, the output gap, uncertainty, and their interaction terms are divided by $1-\rho$, where $\rho \equiv \rho_1 + \rho_2$, and are interpreted as long-run effects. The JLN measure is standardized. Heteroskedasticity-robust standard errors are reported. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.2: Taylor rule using the 12-month JLN measure

	(1) OLS	(2) OLS	(3) GMM	(4) GMM	(5) GMM Shocks	(6) GMM Shocks
π_t^{cpi}	1.46***	1.93***	1.94***	2.23***	0.75***	0.84***
	(0.42)	(0.39)	(0.39)	(0.29)	(0.25)	(0.27)
$\pi_t^{cpi} \times JLN_{t-1}^{12m}$		-1.11**		-1.23**		-0.73**
		(0.56)		(0.58)		(0.30)
y_t	1.34***	1.37***	1.01***	1.05***	1.38***	1.45***
	(0.20)	(0.20)	(0.23)	(0.16)	(0.14)	(0.11)
$y_t \times JLN_{t-1}^{12m}$		-0.35**		0.02		-0.37**
		(0.16)		(0.21)		(0.18)
JLN_{t-1}^{12m}		0.77		2.26		-0.23
		(1.26)		(1.66)		(1.17)
ρ	0.87***	0.87***	0.88***	0.83***	0.89***	0.87***
	(0.04)	(0.04)	(0.05)	(0.06)	(0.01)	(0.02)
N	139	139	139	139	106	106

Notes: This table uses the JLN measure based on 12-month-ahead forecasts. Columns (1), (3), and (5) report estimates of equation (1), and columns (2), (4), and (6) report estimates of equation (2). Columns (1) and (2) report OLS estimates; columns (3) and (4) report GMM estimates using the same instruments as in Clarida, Gali, and Gertler (2000); and columns (5) and (6) report GMM estimates using only lagged macroeconomic shocks as instruments: the business-cycle shock of Angeletos, Collard, and Dellas (2020), the excess bond-premium shock of Gilchrist and Zakrajšek (2012), the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). These shocks enter the instrument set with lags 5 through 20. The coefficients on inflation, the output gap, uncertainty, and their interaction terms are divided by $1-\rho$, where $\rho \equiv \rho_1 + \rho_2$, and are interpreted as long-run effects. The JLN measure is standardized. Heteroskedasticity-robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.3: Taylor rule using VIX as measure of uncertainty

	(1) OLS	(2) OLS	(3) GMM	(4) GMM	(5) GMM Shocks	(6) GMM Shocks
π_t^{cpi}	0.45	0.63	1.27***	1.76***	0.68***	1.33***
	(0.59)	(0.43)	(0.41)	(0.27)	(0.24)	(0.19)
$\pi_t^{cpi} \times VIX_{t-1}$		-1.26**		-1.05***		-1.34***
		(0.50)		(0.35)		(0.25)
y_t	1.28***	1.49***	1.25***	1.27***	1.43***	1.33***
	(0.26)	(0.21)	(0.19)	(0.13)	(0.12)	(0.11)
$y_t \times VIX_{t-1}$		-0.53***		-0.46***		-0.30***
		(0.20)		(0.14)		(0.10)
VIX_{t-1}		0.53		0.72		1.72***
		(1.03)		(1.03)		(0.59)
ρ	0.91***	0.89***	0.88***	0.72***	0.90***	0.88***
	(0.03)	(0.03)	(0.03)	(0.05)	(0.01)	(0.02)
N	115	115	115	115	102	102

Notes: This table uses the VIX as measure of uncertainty. The sample begins in 1990q1 due to data availability. Columns (1), (3), and (5) report estimates of equation (1), and columns (2), (4), and (6) report estimates of equation (2). Columns (1) and (2) report OLS estimates; columns (3) and (4) report GMM estimates using the same instruments as in Clarida, Gali, and Gertler (2000); and columns (5) and (6) report GMM estimates using only lagged macroeconomic shocks as instruments: the business-cycle shock of Angeletos, Collard, and Dellas (2020), the excess bond-premium shock of Gilchrist and Zakrajšek (2012), the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). These shocks enter the instrument set with lags 5 through 20. The coefficients on inflation, the output gap, uncertainty, and their interaction terms are divided by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, and are interpreted as long-run effects. The VIX is standardized. Heteroskedasticity-robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.4: Taylor rule using GDP deflator as measure of inflation

	(1) OLS	(2) OLS	(3) GMM	(4) GMM	(5) GMM Shocks	(6) GMM Shocks
π_t^{gdpd}	1.43**	2.03***	0.63	2.74***	0.56**	0.61*
	(0.66)	(0.70)	(1.12)	(0.62)	(0.26)	(0.36)
$\pi_t^{gdpd} \times JLN_{t-1}$		-0.35		-1.20		0.15
		(0.62)		(0.93)		(0.51)
y_t	1.48***	1.45***	1.48***	1.39***	1.52***	1.45***
	(0.30)	(0.31)	(0.40)	(0.21)	(0.10)	(0.13)
$y_t \times JLN_{t-1}$		-0.82**		-0.65		-0.65***
		(0.41)		(0.61)		(0.24)
JLN_{t-1}		-2.72		-1.86		-3.00**
		(1.89)		(2.28)		(1.40)
ρ	0.91***	0.90***	0.94***	0.85***	0.90***	0.88***
	(0.03)	(0.03)	(0.02)	(0.04)	(0.01)	(0.02)
N	139	139	139	139	106	106

Notes: This table uses GDP deflator inflation. Columns (1), (3), and (5) report estimates of equation (1), and columns (2), (4), and (6) report estimates of equation (2). Columns (1) and (2) report OLS estimates; columns (3) and (4) report GMM estimates using the same instruments as in Clarida, Gali, and Gertler (2000); and columns (5) and (6) report GMM estimates using only lagged macroeconomic shocks as instruments: the business-cycle shock of Angeletos, Collard, and Dellas (2020), the excess bond-premium shock of Gilchrist and Zakrajšek (2012), the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). These shocks enter the instrument set with lags 5 through 20. The coefficients on inflation, the output gap, uncertainty, and their interaction terms are divided by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, and are interpreted as long-run effects. The JLN measure is standardized. Heteroskedasticity-robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.5: Taylor rule using PCE inflation

	(1) OLS	(2) OLS	(3) GMM	(4) GMM	(5) GMM Shocks	(6) GMM Shocks
π_t^{pce}	2.15***	2.36***	2.07***	2.27***	1.21***	1.23***
	(0.39)	(0.32)	(0.32)	(0.29)	(0.20)	(0.29)
$\pi_t^{pce} \times JLN_{t-1}$		-0.34		-0.83**		0.11
		(0.28)		(0.42)		(0.35)
y_t	1.27***	1.31***	1.44***	1.32***	1.47***	1.43***
	(0.16)	(0.14)	(0.14)	(0.14)	(0.07)	(0.08)
$y_t \times JLN_{t-1}$		-0.52***		-0.29		-0.52***
		(0.13)		(0.18)		(0.13)
JLN_{t-1}		-1.19*		-0.46		-2.19**
		(0.67)		(0.87)		(0.92)
ρ	0.84***	0.82***	0.83***	0.81***	0.87***	0.85***
	(0.04)	(0.03)	(0.03)	(0.05)	(0.01)	(0.02)
N	139	139	139	139	106	106

Notes: This table uses PCE inflation. Columns (1), (3), and (5) report estimates of equation (1), and columns (2), (4), and (6) report estimates of equation (2). Columns (1) and (2) report OLS estimates; columns (3) and (4) report GMM estimates using the same instruments as in Clarida, Gali, and Gertler (2000); and columns (5) and (6) report GMM estimates using only lagged macroeconomic shocks as instruments: the business-cycle shock of Angeletos, Collard, and Dellas (2020), the excess bond-premium shock of Gilchrist and Zakrajšek (2012), the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). These shocks enter the instrument set with lags 5 through 20. The coefficients on inflation, the output gap, uncertainty, and their interaction terms are divided by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, and are interpreted as long-run effects. The JLN measure is standardized. Heteroskedasticity-robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.6: Taylor rule without using the shadow rate

	(1) OLS	(2) OLS	(3) GMM	(4) GMM	(5) GMM Shocks	(6) GMM Shocks
π_t^{cpi}	1.38***	1.54***	1.72***	1.73***	0.62***	0.78***
	(0.38)	(0.33)	(0.35)	(0.31)	(0.18)	(0.17)
$\pi_t^{cpi} \times JLN_{t-1}$		-0.78*		-0.72		-0.27
		(0.46)		(0.57)		(0.29)
y_t	1.06***	1.05***	0.74***	0.81***	1.17***	1.11***
	(0.16)	(0.16)	(0.17)	(0.18)	(0.08)	(0.09)
$y_t \times JLN_{t-1}$		-0.34*		-0.07		-0.16
		(0.18)		(0.27)		(0.12)
JLN_{t-1}		-0.18		0.99		-0.03
		(1.14)		(1.69)		(0.92)
ρ	0.85***	0.85***	0.87***	0.86***	0.88***	0.88***
	(0.05)	(0.05)	(0.05)	(0.05)	(0.02)	(0.02)
N	139	139	139	139	106	106

Notes: This table uses the federal funds rate throughout the sample. Columns (1), (3), and (5) report estimates of equation (1), and columns (2), (4), and (6) report estimates of equation (2). Columns (1) and (2) report OLS estimates; columns (3) and (4) report GMM estimates using the same instruments as in Clarida, Gali, and Gertler (2000); and columns (5) and (6) report GMM estimates using only lagged macroeconomic shocks as instruments: the business-cycle shock of Angeletos, Collard, and Dellas (2020), the excess bond-premium shock of Gilchrist and Zakrajšek (2012), the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). These shocks enter the instrument set with lags 5 through 20. The coefficients on inflation, the output gap, uncertainty, and their interaction terms are divided by $1-\rho$, where $\rho \equiv \rho_1 + \rho_2$, and are interpreted as long-run effects. The JLN measure is standardized. Heteroskedasticity-robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.7: Taylor rule controlling for recessions

	(1) OLS	(2) OLS	(3) GMM	(4) GMM	(5) GMM Shocks	(6) GMM Shocks
π_t^{cpi}	1.46***	1.75***	1.94***	2.47***	0.75***	1.08***
	(0.42)	(0.39)	(0.39)	(0.32)	(0.25)	(0.25)
$\pi_t^{cpi} \times JLN_{t-1}$		-1.13**		-1.08**		-0.64*
		(0.51)		(0.46)		(0.35)
y_t	1.34***	1.25***	1.01***	0.98***	1.38***	1.46***
	(0.20)	(0.20)	(0.23)	(0.15)	(0.14)	(0.11)
$y_t \times JLN_{t-1}$		-0.47*		-0.17		-0.41**
		(0.26)		(0.24)		(0.21)
JLN_{t-1}		1.44		2.49*		0.61
		(1.24)		(1.37)		(1.23)
Rec_{t-1}		-5.90**		-3.99		-5.55**
		(2.75)		(5.02)		(2.66)
$\pi_t^{cpi} \times Rec_{t-1}$		1.03		0.25		0.87
		(0.70)		(1.35)		(0.64)
$y_t \times Rec_{t-1}$		0.55		0.23		0.01
		(0.84)		(1.06)		(0.58)
ρ	0.87***	0.84***	0.88***	0.75***	0.89***	0.85***
	(0.04)	(0.05)	(0.05)	(0.08)	(0.01)	(0.03)
N	139	139	139	139	106	106

Notes: This table includes a lagged recession dummy, Rec_{t-1}, and its interactions with inflation and the output gap as additional controls. Columns (1), (3), and (5) report estimates of equation (1), and columns (2), (4), and (6) report estimates of equation (2), both augmented with these controls. Columns (1) and (2) report OLS estimates; columns (3) and (4) report GMM estimates using the same instruments as in Clarida, Gali, and Gertler (2000); and columns (5) and (6) report GMM estimates using only lagged macroeconomic shocks as instruments: the business-cycle shock of Angeletos, Collard, and Dellas (2020), the excess bond-premium shock of Gilchrist and Zakrajšek (2012), the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). These shocks enter the instrument set with lags 5 through 20. The coefficients on inflation, the output gap, uncertainty, the recession dummy, and their interaction terms are divided by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, and are interpreted as long-run effects. The JLN measure is standardized. Heteroskedasticity-robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.8: Taylor rule with uncertainty-dependent interest-rate smoothing

	(1) OLS	(2) GMM	(3) GMM Shocks
Inflation: JLN=-1	6.86	3.94*	0.91
	(7.18)	(2.09)	(0.85)
Inflation: JLN=0	1.63***	2.03***	0.82***
	(0.33)	(0.30)	(0.32)
Inflation: JLN=1	0.92**	1.07**	0.77***
	(0.37)	(0.47)	(0.28)
Output: JLN=-1	3.96	1.26**	2.04***
	(3.51)	(0.51)	(0.39)
Output: JLN=0	1.12***	0.97***	1.29***
	(0.19)	(0.23)	(0.13)
Output: JLN=1	0.74***	0.83***	0.88***
	(0.15)	(0.28)	(0.20)
N	139	139	106

Notes: This table reports conditional long-run responses from equation (10), which allows interest-rate smoothing to vary with uncertainty. The responses to inflation and the output gap are $(\beta_1 + \beta_2 JLN)/[1 - \rho(JLN)]$ and $(\gamma_1 + \gamma_2 JLN)/[1 - \rho(JLN)]$, respectively, where $\rho(JLN) = \rho_1 + \rho_2 + (a_1 + a_2)JLN$. The JLN measure is standardized, so $JLN = -1, 0, 1$ correspond to one standard deviation below the mean, the mean, and one standard deviation above the mean. Column (1) reports OLS estimates; column (2) reports GMM estimates using the same instruments as in Clarida, Gali, and Gertler (2000); and column (3) reports GMM estimates using only lagged macroeconomic shocks as instruments: the business-cycle shock of Angeletos, Collard, and Dellas (2020), the excess bond-premium shock of Gilchrist and Zakrajšek (2012), the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). These shocks enter the instrument set with lags 5 through 20. Standard errors computed using the delta method are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

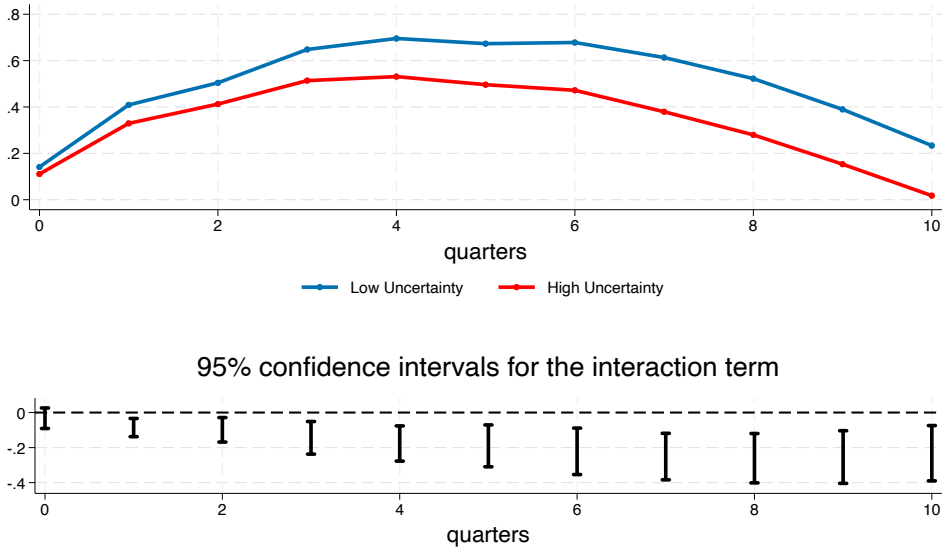
A.2. Robustness of the local projections

This section reports the robustness checks discussed in Section 4. All figures follow the format used in the main text.

Figure A.1 and Figure A.2 report estimates of equation (9) without the controls W^x_{t-1}. The results remain qualitatively similar to the benchmark estimates: under higher uncertainty, the policy rate responds less to a given demand shock, while unemployment declines more and inflation rises more. Thus, the main findings are not driven by the inclusion of the control vector in the benchmark local projections.

Figure A.3 and Figure A.4 report estimates of equation (9) replacing the JLN mea-

Figure A.1: Local projections of the federal funds rate without controls

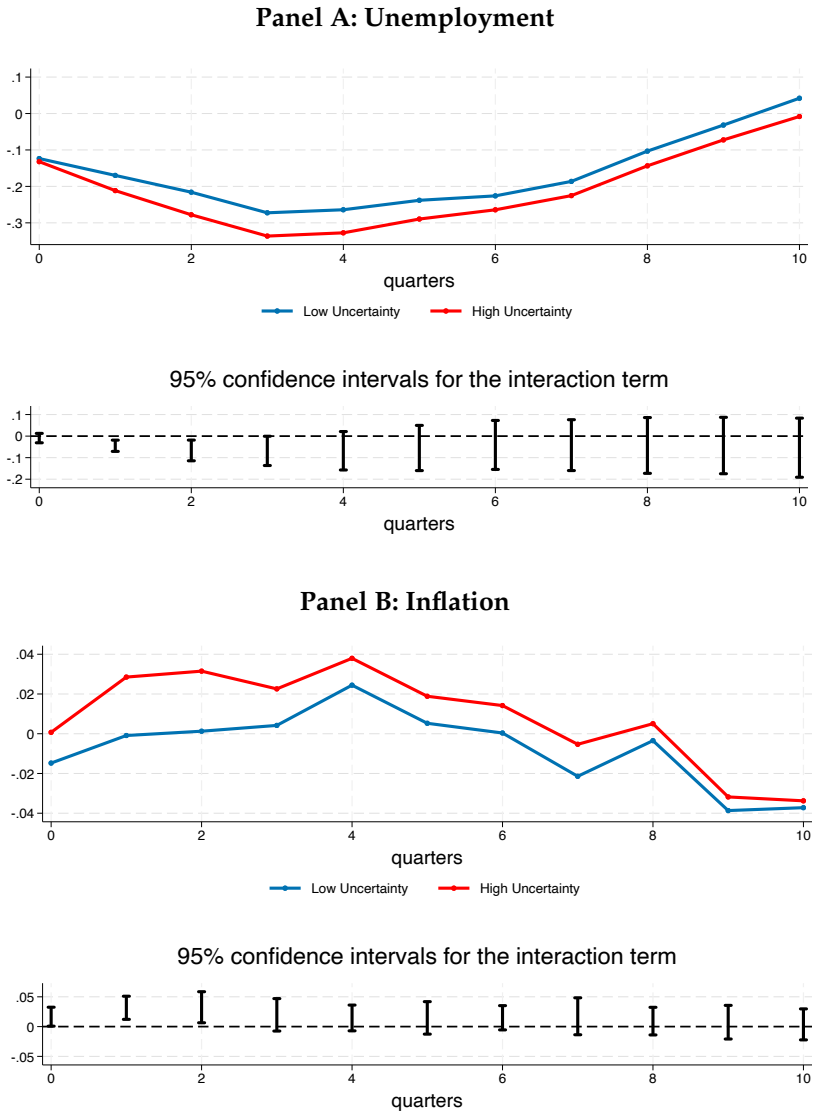


Notes: The figure reports estimates of equation (9) using the federal funds rate as the dependent variable. The top panel plots $\hat{\beta}^i_{1,j}\overline{Shock} + \hat{\beta}^i_{2,j}(\widetilde{JLN} \times \overline{Shock})$ for horizons $j = 0, \ldots, 10$. $\overline{Shock}$ denotes one standard deviation of the business-cycle unemployment shock, and $\widetilde{JLN}$ is set to the 25th percentile of the JLN distribution for the low-uncertainty response and to the 75th percentile for the high-uncertainty response. The bottom panel reports the 95% confidence intervals for $\hat{\beta}^i_{2,j}$.

sure of uncertainty with the VIX. The policy-rate and unemployment responses remain qualitatively similar to the benchmark estimates. For inflation, the difference across uncertainty states is less pronounced, although the response generally remains larger under high uncertainty. Because the VIX is available only from 1990q1, the sample is shorter. Confidence intervals are wider, particularly for unemployment and inflation, but the interaction coefficients remain statistically significant at some horizons.

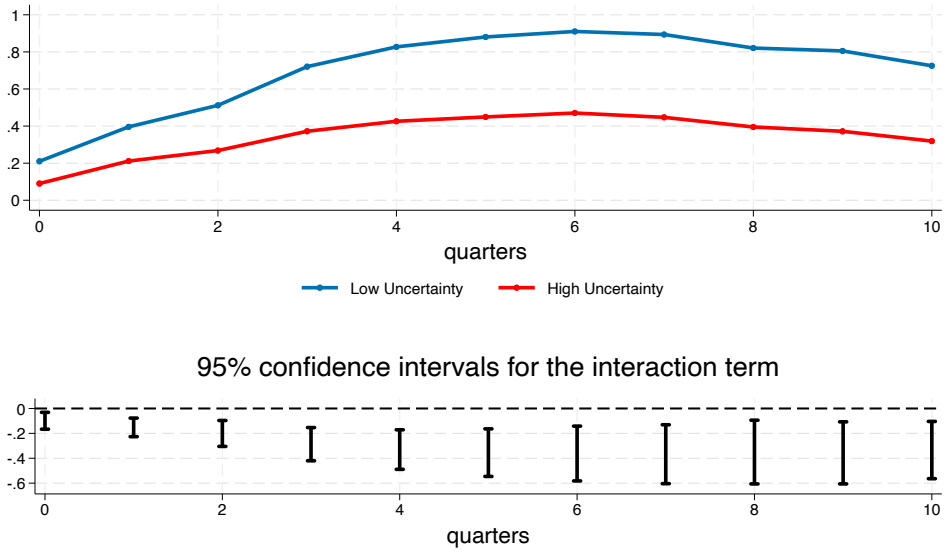
Figure A.5 and Figure A.6 report estimates of equation (9) using the excess bond premium shock of Gilchrist and Zakrajšek (2012) in place of the benchmark business-cycle unemployment shock. I normalize the shock so that a positive realization corresponds to a decline in the excess bond premium and an increase in the policy rate. The results remain qualitatively similar to the benchmark: under higher uncertainty, the policy rate responds less, inflation rises more, and unemployment declines more. The difference in unemployment responses becomes less pronounced at longer horizons.

Figure A.2: Local projections of unemployment and inflation without controls



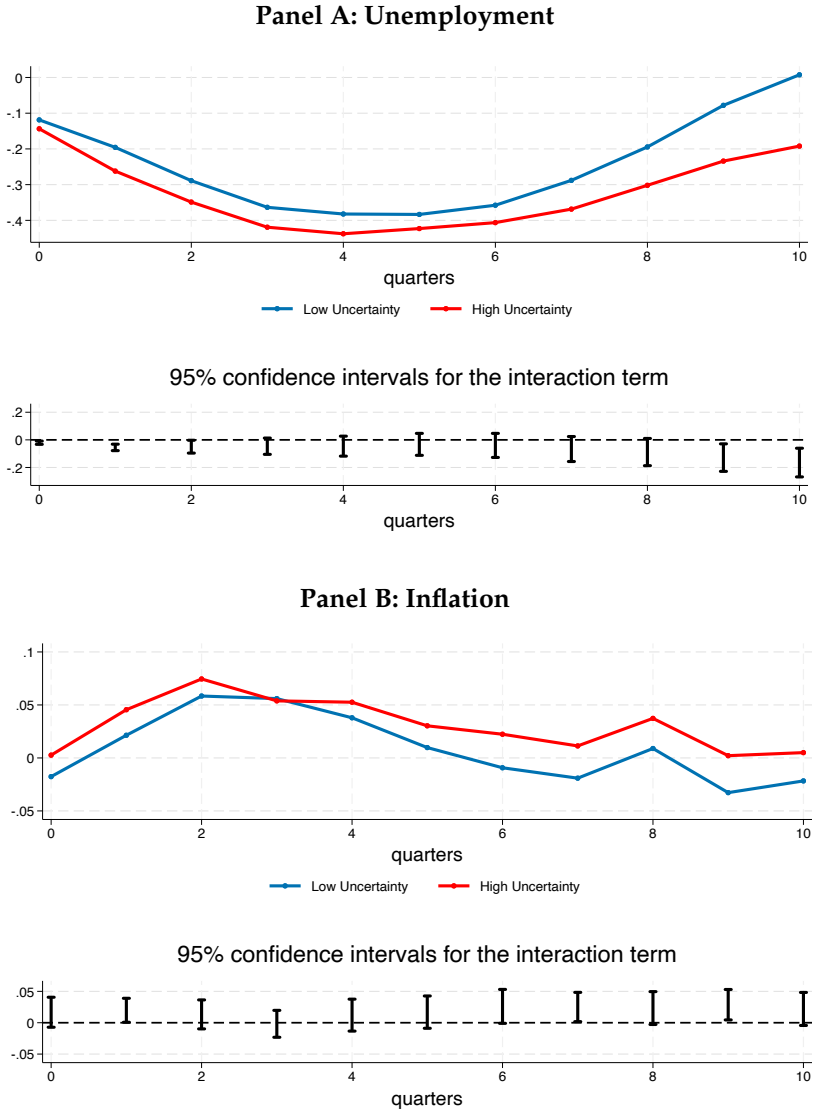
Notes: Panels A and B report estimates of equation (9) excluding the controls W_{t-1}^{x}, using unemployment and inflation as the dependent variables, respectively. Within each panel, the upper plot shows $\hat{\beta}_{1,j}^{x}\overline{Shock} + \hat{\beta}_{2,j}^{x}(\widetilde{JLN} \times \overline{Shock})$ for horizons $j = 0, \ldots, 10$, where $x = u$ in Panel A and $x = \pi$ in Panel B. $\overline{Shock}$ denotes one standard deviation of the business-cycle unemployment shock, and $\widetilde{JLN}$ is set to the 25th percentile of the JLN distribution for the low-uncertainty response and to the 75th percentile for the high-uncertainty response. Within each panel, the lower plot reports the 95% confidence intervals for $\hat{\beta}_{2,j}^{x}$.

Figure A.3: Local projections of the federal funds rate using the VIX



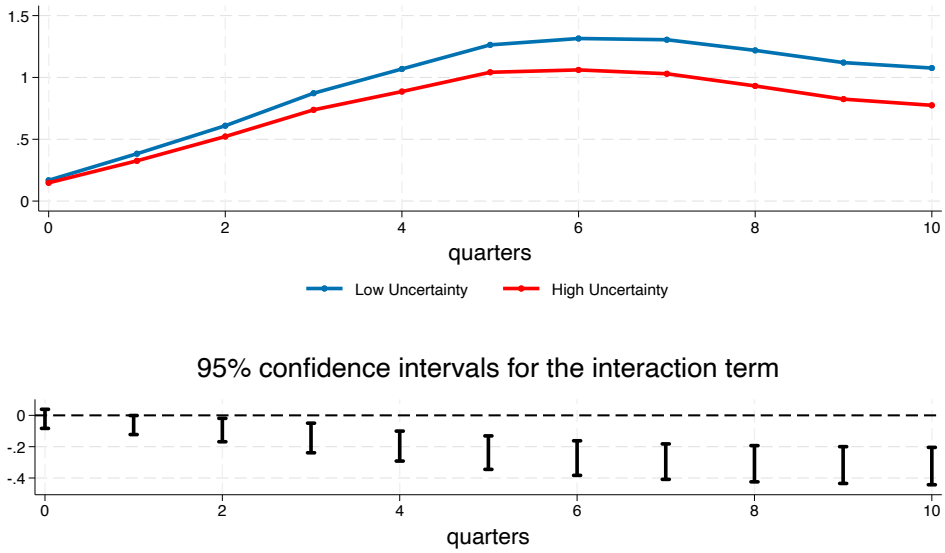
Notes: The figure reports estimates of equation (9) using the federal funds rate as the dependent variable. The top panel plots $\hat{\beta}^i_{1,j}\overline{Shock} + \hat{\beta}^i_{2,j}(\widetilde{VIX} \times \overline{Shock})$ for horizons $j = 0, \ldots, 10$. $\overline{Shock}$ denotes one standard deviation of the business-cycle unemployment shock, and $\widetilde{VIX}$ is set to the 25th percentile of the VIX distribution for the low-uncertainty response and to the 75th percentile for the high-uncertainty response. The bottom panel reports the 95% confidence intervals for $\hat{\beta}^i_{2,j}$.

Figure A.4: Local projections of unemployment and inflation using the VIX



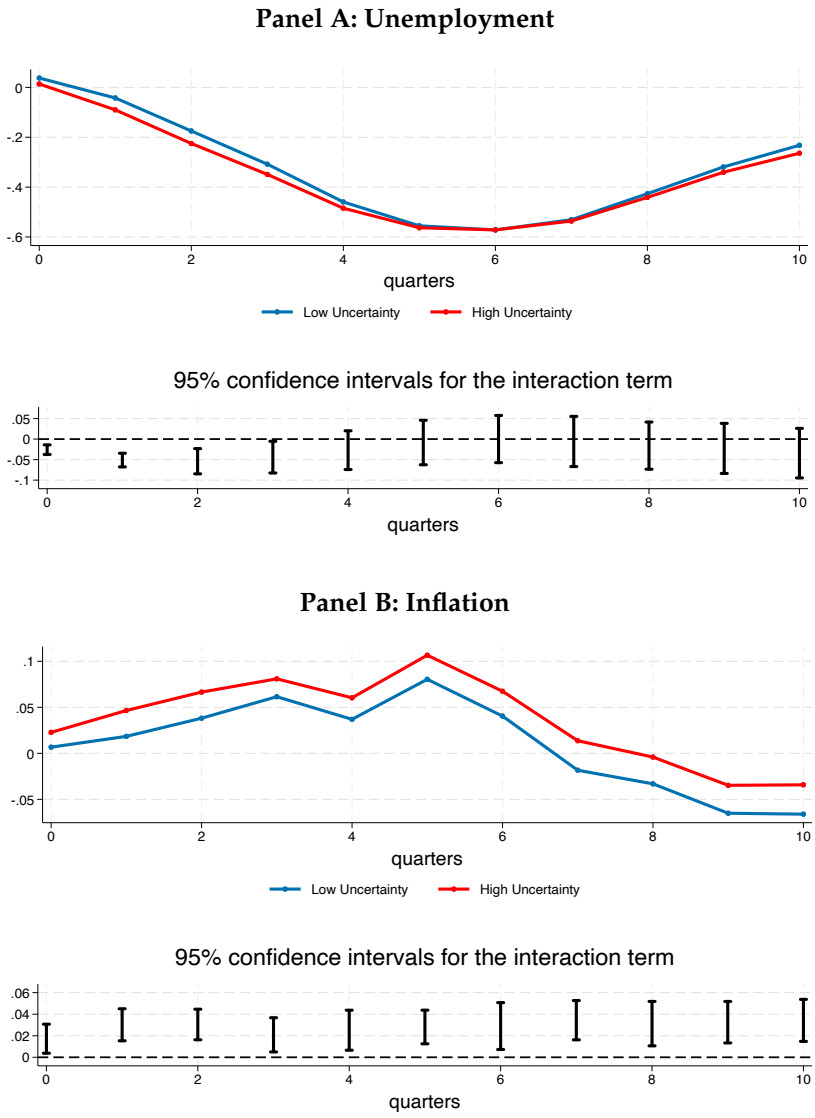
Notes: Panels A and B report estimates of equation (9) replacing the JLN measure of uncertainty with the VIX, using unemployment and inflation as the dependent variables, respectively. Within each panel, the upper plot shows $\hat{\beta}^x_{1,j}\overline{Shock} + \hat{\beta}^x_{2,j}(\widetilde{VIX} \times \overline{Shock})$ for horizons $j = 0, \ldots, 10$, where $x = u$ in Panel A and $x = \pi$ in Panel B. $\overline{Shock}$ denotes one standard deviation of the business-cycle unemployment shock, and $\widetilde{VIX}$ is set to the 25th percentile of the VIX distribution for the low-uncertainty response and to the 75th percentile for the high-uncertainty response. Within each panel, the lower plot reports the 95% confidence intervals for $\hat{\beta}^x_{2,j}$.

Figure A.5: Local projections of the federal funds rate using the excess bond premium shock



Notes: The figure reports estimates of equation (9) using the federal funds rate as the dependent variable. The top panel plots $\hat{\beta}^i_{1,j}\overline{Shock} + \hat{\beta}^i_{2,j}(\widetilde{JLN} \times \overline{Shock})$ for horizons $j = 0, \ldots, 10$. $\overline{Shock}$ denotes one standard deviation of the excess bond premium shock, and $\widetilde{JLN}$ is set to the 25th percentile of the JLN distribution for the low-uncertainty response and to the 75th percentile for the high-uncertainty response. The bottom panel reports the 95% confidence intervals for $\hat{\beta}^i_{2,j}$.

Figure A.6: Local projections of unemployment and inflation using the excess bond premium shock



Notes: Panels A and B report estimates of equation (9) using the excess bond premium shock, with unemployment and inflation as the dependent variables, respectively. Within each panel, the upper plot shows $\hat{\beta}^x_{1,j}\overline{Shock} + \hat{\beta}^x_{2,j}(\widetilde{JLN} \times \overline{Shock})$ for horizons $j = 0, \ldots, 10$, where $x = u$ in Panel A and $x = \pi$ in Panel B. $\overline{Shock}$ denotes one standard deviation of the excess bond premium shock, and $\widetilde{JLN}$ is set to the 25th percentile of the JLN distribution for the low-uncertainty response and to the 75th percentile for the high-uncertainty response. Within each panel, the lower plot reports the 95% confidence intervals for $\hat{\beta}^x_{2,j}$.

B. Model's proofs and additional results

B.1. Proof of Lemma 1

In this section, I show that certainty equivalence holds regardless of the private sector's information set, provided that it is the same as or finer than that of the monetary authority.

I first note that, under full information, the optimal interest rate is

$$i_t^* = \sigma a_t. \tag{11}$$

This policy implements the first-best allocation, under which inflation and the output gap are zero in every period. This is the standard divine coincidence result.

Under partial information, the first-order condition is

$$0 = \mathbb{E}_t^m \left[\kappa \pi_t + \lambda_y y_t \right]. \tag{12}$$

Taking the monetary authority's conditional expectation of equation (3) and equation (4) gives

$$\mathbb{E}_t^m \pi_t = \kappa \mathbb{E}_t^m y_t + \beta \mathbb{E}_t^m \pi_{t+1}, \tag{13}$$

and

$$\mathbb{E}_t^m y_t = \mathbb{E}_t^m y_{t+1} - \frac{1}{\sigma}(i_t - \mathbb{E}_t^m \pi_{t+1}) + \mathbb{E}_t^m a_t. \tag{14}$$

Here I use the fact that i_t is observed by the monetary authority and that

$$\mathbb{E}_t^m \mathbb{E}_t^p x = \mathbb{E}_t^m x$$

for any integrable random variable x, because the private sector's information set is the same as or finer than that of the monetary authority.

Substituting equation (14) and equation (13) into equation (12) yields

$$i_t^* = \sigma(\mathbb{E}_t^m y_{t+1} + \mathbb{E}_t^m a_t) + \left[1 + \frac{\sigma \kappa \beta}{\kappa^2 + \lambda_y} \right] \mathbb{E}_t^m \pi_{t+1}. \tag{15}$$

Thus, certainty equivalence follows if $\mathbb{E}_t^m y_{t+1} = 0$ and $\mathbb{E}_t^m \pi_{t+1} = 0$.

To establish these conditions, consider any period τ. Combining equation (13) with equation (12) gives

$$\mathbb{E}_\tau^m \pi_\tau = \alpha \mathbb{E}_\tau^m \pi_{\tau+1}, \qquad \alpha \equiv \frac{\beta \lambda_y}{\kappa^2 + \lambda_y} < 1. \tag{16}$$

Iterating forward,

$$\mathbb{E}_\tau^m \pi_\tau = \alpha^n \mathbb{E}_\tau^m \pi_{\tau+n}. \tag{17}$$

A bounded equilibrium implies

$$\lim_{n\to\infty} \alpha^n \mathbb{E}_\tau^m \pi_{\tau+n} = 0.$$

Therefore,

$$\mathbb{E}_\tau^m \pi_\tau = 0.$$

Setting $\tau = t+1$ and taking the monetary authority's conditional expectation at time t then gives

$$\mathbb{E}_t^m \pi_{t+1} = 0. \tag{18}$$

Using equation (12) at $t+1$, together with equation (13), and taking the monetary authority's conditional expectation at time t similarly yields

$$\mathbb{E}_t^m y_{t+1} = -\frac{\kappa\beta}{\kappa^2+\lambda_y} \mathbb{E}_t^m \pi_{t+2} = 0. \tag{19}$$

Substituting these results into the optimality condition for the interest rate gives

$$i_t^* = \sigma \mathbb{E}_t^m a_t. \tag{20}$$

Hence, certainty equivalence holds.

Intuitively, conditional on its information, the monetary authority chooses the interest rate to offset its estimate of the demand shock. As a result, it expects both inflation and the output gap to be zero.

B.2. Proof of Proposition 1

Because the model contains no backward-looking endogenous variables and d_t is serially independent with mean zero, the continuation terms vanish under both private-sector and monetary-authority expectations:

$$\mathbb{E}_t^p y_{t+1} = \mathbb{E}_t^p \pi_{t+1} = 0, \qquad \mathbb{E}_t^m y_{t+1} = \mathbb{E}_t^m \pi_{t+1} = 0.$$

Using equation (3) and equation (4), I can therefore write the date-t equilibrium conditions as

$$y_t = a_t - \frac{1}{\sigma} i_t, \qquad \pi_t = \kappa y_t.$$

Substituting these expressions into the monetary authority's period loss and conditioning on $\mathcal{I}_t^m$, the first-order condition gives

$$i_t^* = \sigma \mathbb{E}_t^m[a_t].$$

I define

$$q_{t|t} \equiv q(r_t, \omega_t) \equiv \Pr(s_t = D \mid \mathcal{I}_t^m)$$

as the monetary authority's posterior probability that the demand regime is active at time t.[6] Since $a_t = d_t$ in regime D and $a_t = 0$ in regime N, I can write the optimal interest rate as

$$i_t^* = \sigma q_{t|t} \mathbb{E}_t^m[d_t \mid s_t = D].$$

I next derive the posterior probability $q_{t|t}$ and the conditional mean $\mathbb{E}_t^m[d_t \mid s_t = D]$.

The observed signals depend on the contemporaneous policy rate:

$$\widetilde{y}_t = a_t - \frac{1}{\sigma} i_t + v_t^y, \qquad \widetilde{\pi}_t = \kappa \left(a_t - \frac{1}{\sigma} i_t \right) + v_t^\pi.$$

As in the main text, I define the policy-adjusted signals as

$$r_t^y \equiv \widetilde{y}_t + \frac{1}{\sigma} i_t, \qquad r_t^\pi \equiv \widetilde{\pi}_t + \frac{\kappa}{\sigma} i_t.$$

Since the monetary authority observes the raw signals and the policy rate, both adjusted signals belong to $\mathcal{I}_t^m$. Therefore,

$$r_t^y = a_t + v_t^y, \qquad r_t^\pi = \kappa a_t + v_t^\pi.$$

I collect the adjusted signals in

$$r_t \equiv \begin{bmatrix} r_t^y \\ r_t^\pi \end{bmatrix}, \qquad h \equiv \begin{bmatrix} 1 \\ \kappa \end{bmatrix}, \qquad H \equiv h'h = 1 + \kappa^2, \qquad M \equiv \chi + H.$$

Then,

$$r_t = h a_t + v_t, \qquad v_t \sim N(0, \chi \omega_t I_2).$$

The realization of ω_t is observed by both private agents and the monetary authority.

[6] To simplify notation, I suppress the dependence of $q(r_t, \omega_t)$ on the predetermined prior $q_{t|t-1}$.

Conditional on the active regime,

$$r_t \mid D \sim N(0, \Sigma_D(\omega_t)), \qquad \Sigma_D(\omega_t) = \omega_t\big(hh' + \chi I_2\big).$$

Conditional on the inactive regime,

$$r_t \mid N \sim N(0, \Sigma_N(\omega_t)), \qquad \Sigma_N(\omega_t) = \chi\omega_t I_2.$$

Applying Bayes' rule after observing the adjusted signal r_t, I obtain

$$q_{t|t} = \frac{q_{t|t-1} L_D(r_t;\omega_t)}{q_{t|t-1} L_D(r_t;\omega_t) + (1 - q_{t|t-1}) L_N(r_t;\omega_t)}.$$

Here, $L_j(r_t;\omega_t)$ denotes the likelihood of r_t in regime $j \in \{D, N\}$, and

$$q_{t|t-1} \equiv \Pr(s_t = D \mid \mathcal{I}^m_{t-1}) = p_{DD} q_{t-1|t-1} + (1 - p_{NN})(1 - q_{t-1|t-1})$$

is the prior probability implied by the Markov transition matrix.

Equivalently, I define the posterior log-odds as

$$\ell_{t|t} \equiv \log \frac{q_{t|t}}{1 - q_{t|t}}.$$

Using Bayes' rule,

$$\ell_{t|t} = \log \frac{q_{t|t-1}}{1 - q_{t|t-1}} + \log \frac{L_D(r_t;\omega_t)}{L_N(r_t;\omega_t)}.$$

To compute the likelihood ratio, I use the matrix determinant lemma,

$$\det(A + uv') = \det(A)\Big(1 + v' A^{-1} u\Big).$$

Setting $A = \chi I_2$ and $u = v = h$ gives

$$\frac{\det \Sigma_D(\omega_t)}{\det \Sigma_N(\omega_t)} = \frac{\det(\chi I_2 + hh')}{\det(\chi I_2)} = 1 + \frac{H}{\chi}.$$

I also use the Sherman–Morrison formula,

$$(A + uv')^{-1} = A^{-1} - \frac{A^{-1} u v' A^{-1}}{1 + v' A^{-1} u}.$$

Again setting $A = \chi I_2$ and $u = v = h$ gives

$$\left(hh' + \chi I_2\right)^{-1} - (\chi I_2)^{-1} = -\frac{hh'}{\chi(\chi + H)}.$$

Combining these two identities, I obtain the Gaussian likelihood ratio

$$\log \frac{L_D(r_t;\omega_t)}{L_N(r_t;\omega_t)} = -\frac{1}{2}\log\left(1 + \frac{H}{\chi}\right) + \frac{(h'r_t)^2}{2\omega_t \chi M}.$$

Hence, the relevant statistic for identifying the regime is the projection of the adjusted signal onto h:

$$S_t \equiv h'r_t = r_t^y + \kappa r_t^\pi.$$

The component of r_t orthogonal to h contains only measurement noise and therefore provides no information about whether the demand regime is active.

Holding r_t and the prior $q_{t|t-1}$ fixed, the derivative of the posterior log-odds with respect to ω_t is

$$\frac{\partial \ell_{t|t}}{\partial \omega_t} = -\frac{S_t^2}{2\omega_t^2 \chi M} \leq 0.$$

Since

$$q_{t|t} = \frac{\exp(\ell_{t|t})}{1 + \exp(\ell_{t|t})},$$

I obtain

$$\frac{\partial q_{t|t}}{\partial \omega_t} = q_{t|t}(1 - q_{t|t})\frac{\partial \ell_{t|t}}{\partial \omega_t} \leq 0,$$

with a strict inequality whenever $S_t \neq 0$ and $q_{t|t} \in (0,1)$. Therefore, for a given policy-adjusted signal r_t, higher uncertainty reduces the posterior probability that the demand regime is active.

Conditional on the active regime, the monetary authority's posterior mean of d_t is

$$\mathbb{E}_t^m[d_t \mid s_t = D] = \omega_t h' \left[\omega_t\left(hh' + \chi I_2\right)\right]^{-1} r_t.$$

The common scale factor ω_t cancels, so

$$\mathbb{E}_t^m[d_t \mid s_t = D] = \frac{h'r_t}{M} = \frac{r_t^y + \kappa r_t^\pi}{\chi + 1 + \kappa^2}.$$

Since $a_t = 0$ in the inactive regime, I obtain

$$\mathbb{E}_t^m[a_t] = q_{t|t}\frac{S_t}{M}.$$

Because the policy adjustment removes the contemporaneous effect of the interest rate from the signals, r_t does not depend on i_t. Hence, the posterior $q_{t|t}$ is determined by the adjusted signal r_t and uncertainty ω_t. Using the expression for $\mathbb{E}_t^m[a_t]$ derived above, the optimal interest rate is

$$i_t^* = \sigma q_{t|t}\frac{S_t}{M}.$$

From the definitions of the adjusted signals,

$$S_t = \widetilde{y}_t + \kappa\widetilde{\pi}_t + \frac{H}{\sigma}i_t^*.$$

Substituting this expression into the optimal policy rule and solving for i_t^* gives

$$i_t^* = \frac{\sigma q_{t|t}}{M - Hq_{t|t}}(\widetilde{y}_t + \kappa\widetilde{\pi}_t).$$

Since $q_{t|t} \equiv q(r_t, \omega_t)$ and $M = \chi + H$, I can write

$$i_t^* = \phi_\pi(\omega_t)\widetilde{\pi}_t + \phi_y(\omega_t)\widetilde{y}_t,$$

where

$$\phi_y(\omega_t) = \frac{\sigma q(r_t, \omega_t)}{\chi + H - Hq(r_t, \omega_t)}, \qquad \phi_\pi(\omega_t) = \kappa\phi_y(\omega_t).$$

The denominator is strictly positive because

$$\chi + H - Hq(r_t, \omega_t) = \chi + H[1 - q(r_t, \omega_t)] > 0.$$

Finally, $\phi_y(\omega_t)$ is increasing in $q_{t|t}$, and so is $\phi_\pi(\omega_t)$ because $\kappa > 0$. Since $q_{t|t}$ is decreasing in ω_t for a given policy-adjusted signal r_t, both policy coefficients are decreasing in uncertainty:

$$\frac{\partial\phi_y(\omega_t)}{\partial\omega_t} < 0, \qquad \frac{\partial\phi_\pi(\omega_t)}{\partial\omega_t} < 0.$$

B.3. Proof of Corollary 1

From Proposition 1, the optimal interest rate is

$$i_t^* = \phi_\pi(\omega_t)\widetilde{\pi}_t + \phi_y(\omega_t)\widetilde{y}_t.$$

For a given policy-adjusted signal r_t, the policy rule is linear in the observed inflation and output-gap signals for any fixed value of ω_t. Therefore, to obtain Corollary 1, I only need to approximate the two policy coefficients to first order with respect to ω_t.

Let $\bar{\omega} > 0$ denote the point around which I expand the coefficient functions, and define

$$\Delta\omega_t \equiv \omega_t - \bar{\omega}.$$

For $j \in \{\pi, y\}$, a first-order Taylor expansion gives

$$\phi_j(\omega_t) = \phi_j(\bar{\omega}) + \phi_j'(\bar{\omega})\Delta\omega_t + o(|\Delta\omega_t|).$$

Substituting these expansions into the optimal policy rule gives

$$\begin{aligned} i_t^* = {} & \left[\phi_\pi(\bar{\omega}) + \phi_\pi'(\bar{\omega})\Delta\omega_t\right]\widetilde{\pi}_t + \left[\phi_y(\bar{\omega}) + \phi_y'(\bar{\omega})\Delta\omega_t\right]\widetilde{y}_t \\ & + o(|\Delta\omega_t|)(|\widetilde{\pi}_t| + |\widetilde{y}_t|). \end{aligned}$$

Using $\Delta\omega_t = \omega_t - \bar{\omega}$, I can rewrite the leading terms as

$$\begin{aligned} i_t^* \approx {} & \left[\phi_\pi(\bar{\omega}) - \bar{\omega}\phi_\pi'(\bar{\omega})\right]\widetilde{\pi}_t + \phi_\pi'(\bar{\omega})(\widetilde{\pi}_t\omega_t) \\ & + \left[\phi_y(\bar{\omega}) - \bar{\omega}\phi_y'(\bar{\omega})\right]\widetilde{y}_t + \phi_y'(\bar{\omega})(\widetilde{y}_t\omega_t). \end{aligned}$$

Therefore, the coefficients in Corollary 1 are

$$\beta_\pi \equiv \phi_\pi(\bar{\omega}) - \bar{\omega}\phi_\pi'(\bar{\omega}), \qquad \gamma_\pi \equiv \phi_\pi'(\bar{\omega}),$$

and

$$\beta_y \equiv \phi_y(\bar{\omega}) - \bar{\omega}\phi_y'(\bar{\omega}), \qquad \gamma_y \equiv \phi_y'(\bar{\omega}).$$

For a given policy-adjusted signal with $S_t \neq 0$, Proposition 1 implies

$$\phi_y(\bar{\omega}) > 0, \qquad \phi_\pi(\bar{\omega}) > 0, \qquad \phi_y'(\bar{\omega}) < 0, \qquad \phi_\pi'(\bar{\omega}) < 0.$$

The signs of $\phi_\pi(\bar{\omega})$ and $\phi'_\pi(\bar{\omega})$ use $\kappa > 0$. Since $\bar{\omega} > 0$, it follows that

$$\beta_y > 0, \qquad \beta_\pi > 0, \qquad \gamma_y < 0, \qquad \gamma_\pi < 0.$$

B.4. Proof of Proposition 2

Let $M \equiv \chi + H$. From the proof of Proposition 1, I can write the optimal interest rate in terms of the sufficient statistic $S_t = h' r_t$ as

$$i_t^* = \sigma q(r_t, \omega_t) \frac{S_t}{M}.$$

Since

$$S_t = r_t^y + \kappa r_t^\pi = H a_t + v_t^y + \kappa v_t^\pi,$$

the optimal policy rule becomes

$$i_t^* = \sigma \frac{q(r_t, \omega_t)}{M} \left(H a_t + v_t^y + \kappa v_t^\pi \right).$$

Using the static output equation from the proof of Proposition 1,

$$y_t = a_t - \frac{1}{\sigma} i_t,$$

I obtain

$$y_t^* = a_t - \frac{q(r_t, \omega_t)}{M} \left(H a_t + v_t^y + \kappa v_t^\pi \right).$$

Collecting terms,

$$y_t^* = \left[1 - \frac{H}{M} q(r_t, \omega_t) \right] a_t - \frac{q(r_t, \omega_t)}{M} \left(v_t^y + \kappa v_t^\pi \right).$$

Therefore, the coefficients in Proposition 2 are

$$\gamma_a(\omega_t) \equiv 1 - \frac{H}{M} q(r_t, \omega_t) = 1 - \frac{H}{\chi + H} q(r_t, \omega_t),$$

and

$$\gamma_v(\omega_t) \equiv \frac{q(r_t, \omega_t)}{M} = \frac{q(r_t, \omega_t)}{\chi + H}.$$

Hence,

$$y_t^* = \gamma_a(\omega_t) a_t - \gamma_v(\omega_t) \left(v_t^y + \kappa v_t^\pi \right).$$

The static Phillips curve then gives

$$\pi_t^* = \kappa y_t^*.$$

It remains to establish the comparative statics. From Proposition 1, for a given policy-adjusted signal r_t,

$$\frac{\partial q(r_t, \omega_t)}{\partial \omega_t} < 0$$

whenever $S_t \neq 0$. Since χ and H are fixed parameters,

$$\frac{\partial \gamma_a(\omega_t)}{\partial \omega_t} = -\frac{H}{\chi + H} \frac{\partial q(r_t, \omega_t)}{\partial \omega_t} > 0,$$

and

$$\frac{\partial \gamma_\nu(\omega_t)}{\partial \omega_t} = \frac{1}{\chi + H} \frac{\partial q(r_t, \omega_t)}{\partial \omega_t} < 0.$$

Thus, for a given policy-adjusted signal, higher uncertainty increases the propagation of demand shocks to inflation and the output gap, while reducing the propagation of measurement errors.

B.5. Solution of persistent demand shock

This subsection describes the solution of the model with persistent demand shocks introduced in Section 3.1.2, for a given level of uncertainty, $\omega_t = \omega$ for all t. I consider two information structures. In Section B.5.1, private agents and the monetary authority share the same information set, whereas in Section B.5.4, the private sector has full information.

I first establish a result that applies independently of the private sector's information set. By Lemma 1, optimal monetary policy under discretion satisfies the certainty-equivalence rule

$$i_t^* = \sigma \mathbb{E}_t^m a_t.$$

Therefore, solving the model requires characterizing $\mathbb{E}_t^m a_t$.

I define

$$\mu_{t|t} \equiv \mu_t \equiv \mathbb{E}_t^m a_t = \mathbb{E}[a_t \mid \mathcal{I}_t^m], \qquad \mu_{t|t-1} \equiv \mathbb{E}_{t-1}^m a_t = \mathbb{E}\left[a_t \mid \mathcal{I}_{t-1}^m\right],$$

$$q_{t|t} \equiv q_t \equiv \Pr(s_t = D \mid \mathcal{I}_t^m), \qquad q_{t|t-1} \equiv \Pr(s_t = D \mid \mathcal{I}_{t-1}^m),$$

$$m_{t|t} \equiv m_t \equiv \mathbb{E}[d_t \mid \mathcal{I}_t^m, s_t = D], \qquad m_{t|t-1} \equiv \mathbb{E}\left[d_t \mid \mathcal{I}_{t-1}^m, s_t = D\right],$$

and

$$f_{t|t}(d) \equiv p(d_t = d \mid s_t = D, \mathcal{I}_t^m), \qquad f_{t|t-1}(d) \equiv p(d_t = d \mid s_t = D, \mathcal{I}_{t-1}^m),$$

where $p(\cdot)$ denotes the corresponding density function.

Since $a_t = d_t$ in regime D and $a_t = 0$ in regime N,

$$\mu_t = q_t m_t.$$

Therefore, the certainty-equivalence rule in Lemma 1 can be written as

$$i_t^* = \sigma \mathbb{E}_t^m a_t = \sigma \mu_t = \sigma q_t m_t.$$

Given the priors $q_{t|t-1}$ and $f_{t|t-1}$, solving the filtering problem requires updating q_t and the conditional density $f_{t|t}$, from which m_t is obtained.

I first derive q_t. Consider the signal

$$z_t = a_t + \eta_t, \qquad \eta_t \sim N\left(0, \chi_z \omega \bar{\sigma}_\varepsilon^2\right).$$

After observing z_t, I define the likelihood of the signal in the active regime as

$$\ell_D(z_t) = \int \varphi\left(z_t; d, \chi_z \omega \bar{\sigma}_\varepsilon^2\right) f_{t|t-1}(d)\, dd,$$

and the likelihood in the inactive regime as

$$\ell_N(z_t) = \varphi\left(z_t; 0, \chi_z \omega \bar{\sigma}_\varepsilon^2\right),$$

where $\varphi(x; m, V)$ denotes the normal density with mean m and variance V.

Applying Bayes' rule, I obtain

$$q_t = \frac{q_{t|t-1} \ell_D(z_t)}{q_{t|t-1} \ell_D(z_t) + (1 - q_{t|t-1}) \ell_N(z_t)}.$$

The posterior conditional density of d_t in the active regime is

$$f_{t|t}(d) = \frac{\varphi\left(z_t; d, \chi_z \omega \bar{\sigma}_\varepsilon^2\right) f_{t|t-1}(d)}{\ell_D(z_t)}.$$

The corresponding conditional mean is

$$m_t = \int d\, f_{t|t}(d)\, dd.$$

I next derive the one-step-ahead priors $q_{t+1|t}$ and $f_{t+1|t}$. By the law of motion for the regime,

$$q_{t+1|t} = p_{DD} q_t + p_{DN}(1 - q_t).$$

Here, p_{ij} denotes the probability of transitioning from regime j to regime i.

To derive $f_{t+1|t}$, I define the conditional probabilities

$$\begin{aligned}
\lambda_{DD,t} &\equiv \Pr(s_t = D \mid s_{t+1} = D, \mathcal{I}_t^m) \\
&= \frac{\Pr(s_{t+1} = D \mid s_t = D, \mathcal{I}_t^m) \Pr(s_t = D \mid \mathcal{I}_t^m)}{\Pr(s_{t+1} = D \mid \mathcal{I}_t^m)} \\
&= \frac{p_{DD} q_t}{q_{t+1|t}}, \\
\lambda_{DN,t} &\equiv \Pr(s_t = N \mid s_{t+1} = D, \mathcal{I}_t^m) \\
&= \frac{\Pr(s_{t+1} = D \mid s_t = N, \mathcal{I}_t^m) \Pr(s_t = N \mid \mathcal{I}_t^m)}{\Pr(s_{t+1} = D \mid \mathcal{I}_t^m)} \\
&= \frac{p_{DN}(1 - q_t)}{q_{t+1|t}}.
\end{aligned}$$

These probabilities are the conditional weights on the two possible transitions into regime D at time $t+1$. Therefore, the conditional prior density of d_{t+1} is

$$f_{t+1|t}(d) = \lambda_{DD,t} \int \varphi\Big(d; \rho u, \omega \bar{\sigma}_\varepsilon^2\Big) f_{t|t}(u)\, du + \lambda_{DN,t} \varphi\Big(d; 0, \omega \bar{\sigma}_\varepsilon^2\Big).$$

Computational implementation of the filtering problem I now describe the numerical implementation of the density recursion derived above. I represent the conditional density of the demand shock as a mixture of Gaussian distributions and update each component separately. Let $j \geq 1$ denote the number of consecutive periods spent in regime D during the current demand episode, including the current period. Conditional on j, the date of the most recent transition from N to D is known, and the demand process and signal are linear and Gaussian. Therefore, each component can be updated using a Kalman filter. The index j does not affect the transition probabilities or introduce an additional reset within regime D; it only distinguishes the conditional distributions of the current demand shock associated with different episode durations.

For each component, I define

$$
\begin{aligned}
w_{j,t|t} &\equiv \Pr(s_t = D, \text{ current demand episode has duration } j \mid \mathcal{I}_t^m),\\
m_{j,t|t} &\equiv \mathbb{E}[d_t \mid s_t = D, \text{ duration } j, \mathcal{I}_t^m],\\
P_{j,t|t} &\equiv \operatorname{Var}(d_t \mid s_t = D, \text{ duration } j, \mathcal{I}_t^m).
\end{aligned}
$$

I define $w_{j,t|t-1}$, $m_{j,t|t-1}$, and $P_{j,t|t-1}$ analogously conditional on $\mathcal{I}_{t-1}^m$.

The weights $w_{j,t|t}$ include the probability of being in regime D and therefore sum to q_t, rather than to one. Hence,

$$
q_t = \sum_j w_{j,t|t}, \qquad f_{t|t}(d) = \sum_j \frac{w_{j,t|t}}{q_t} \varphi\Big(d; m_{j,t|t}, P_{j,t|t}\Big).
$$

The probability of regime N is $1 - q_t$. I do not need to retain a distribution of d_t conditional on regime N, because the effective demand wedge is zero in that regime and a new demand shock is drawn upon the next transition from N to D.

Step 1: Initialize the stationary prior without past signals.

I initialize the filter immediately before observing z_0, assuming that the monetary authority has not observed any previous signals. For $0 < p_{DD}, p_{NN} < 1$ and $|\rho| < 1$, the stationary probability of regime D is

$$
q_{0|-1} = \frac{p_{DN}}{p_{DN} + p_{ND}}.
$$

Conditional on regime D, the stationary probability that the current demand episode has duration j is

$$
\alpha_j = (1 - p_{DD}) p_{DD}^{j-1}, \qquad j = 1, 2, \ldots.
$$

Since a new demand episode begins with a mean-zero innovation and subsequently follows the AR(1) process, the initial component moments are

$$
m_{j,0|-1} = 0, \qquad P_{j,0|-1} = \omega \bar{\sigma}_\varepsilon^2 \sum_{h=0}^{j-1} \rho^{2h} = \omega \bar{\sigma}_\varepsilon^2 \frac{1 - \rho^{2j}}{1 - \rho^2}.
$$

The exact stationary mixture has weights

$$
w_{j,0|-1} = q_{0|-1} \alpha_j.
$$

For the numerical implementation, I retain the first J components and renormalize their

weights conditional on regime D:

$$w_{j,0|-1} = q_{0|-1}\frac{(1-p_{DD})p_{DD}^{j-1}}{1-p_{DD}^{J}}, \qquad j = 1, \ldots, J.$$

This normalization preserves the prior probability $q_{0|-1}$ of regime D and, therefore, the probability $1 - q_{0|-1}$ of regime N.

Step 2: Calculate the signal likelihoods.

Given the priors at time t, after observing z_t I calculate the likelihood of the signal within each demand component:

$$\ell_{j,t}(z_t) = \varphi\Big(z_t; m_{j,t|t-1}, P_{j,t|t-1} + \chi_z \omega \bar{\sigma}_\varepsilon^2\Big).$$

The likelihood in regime N is

$$\ell_N(z_t) = \varphi\Big(z_t; 0, \chi_z \omega \bar{\sigma}_\varepsilon^2\Big).$$

Therefore, the likelihood conditional on regime D and the predictive density of the signal are

$$\ell_D(z_t) = \sum_j \frac{w_{j,t|t-1}}{q_{t|t-1}}\ell_{j,t}(z_t),$$
$$p(z_t \mid \mathcal{I}_{t-1}^m) \equiv Z_t = \sum_j w_{j,t|t-1}\ell_{j,t}(z_t) + (1 - q_{t|t-1})\ell_N(z_t).$$

The first expression evaluates the density integral defining $\ell_D(z_t)$ above, while Z_t is the total predictive density of the observed signal.

Step 3: Update the component probabilities.

Applying Bayes' rule, I obtain

$$w_{j,t|t} = \frac{w_{j,t|t-1}\ell_{j,t}(z_t)}{Z_t}, \qquad q_t = \sum_j w_{j,t|t}, \qquad 1 - q_t = \frac{(1 - q_{t|t-1})\ell_N(z_t)}{Z_t}.$$

The signal therefore updates both the posterior probability of regime D and the relative probabilities of the demand components conditional on regime D.

Step 4: Update demand within each component.

Conditional on component j, the Kalman gain and posterior moments are

$$\begin{aligned} K_{j,t} &= \frac{P_{j,t|t-1}}{P_{j,t|t-1} + \chi_z \omega \bar{\sigma}_\varepsilon^2}, \\ m_{j,t|t} &= m_{j,t|t-1} + K_{j,t}\Big(z_t - m_{j,t|t-1}\Big), \\ P_{j,t|t} &= (1 - K_{j,t}) P_{j,t|t-1}. \end{aligned}$$

Conditional on each component, the posterior distribution remains Gaussian. The aggregate posterior remains a Gaussian mixture, with the component probabilities updated in Step 3.

Step 5: Calculate the posterior mean and optimal interest rate.

The conditional mean of the demand shock in regime D and the posterior mean of the effective demand wedge are

$$m_t = \sum_j \frac{w_{j,t|t}}{q_t} m_{j,t|t}, \qquad \mu_t = q_t m_t = \sum_j w_{j,t|t} m_{j,t|t}.$$

Therefore, the optimal interest rate is

$$i_t^* = \sigma \mu_t.$$

Step 6: Predict the next-period components.

If the economy transitions from regime N to regime D, a new demand episode begins. The predicted weight and moments of the new component are

$$w_{1,t+1|t} = p_{DN}(1 - q_t), \qquad m_{1,t+1|t} = 0, \qquad P_{1,t+1|t} = \omega \bar{\sigma}_\varepsilon^2.$$

If the current demand episode continues, each component of duration j becomes a component of duration $j + 1$:

$$\begin{aligned} w_{j+1,t+1|t} &= p_{DD} w_{j,t|t}, \\ m_{j+1,t+1|t} &= \rho m_{j,t|t}, \\ P_{j+1,t+1|t} &= \rho^2 P_{j,t|t} + \omega \bar{\sigma}_\varepsilon^2. \end{aligned}$$

Summing the predicted component weights yields

$$q_{t+1|t} = p_{DD} q_t + p_{DN}(1 - q_t), \qquad 1 - q_{t+1|t} = p_{ND} q_t + p_{NN}(1 - q_t).$$

After observing z_{t+1}, the filter returns to Step 2.

Starting with J components before observing z_0, the filter contains at most $J + t$ components before observing z_t. Thus, there is no repeated truncation, and the filtering recursion is exact conditional on the finite initial mixture, up to numerical rounding.

B.5.1. Same information set

Suppose that the private sector and the monetary authority share the same information set. Since

$$\mu_{t+1} = \mathbb{E}^m_{t+1} a_{t+1},$$

the law of iterated expectations implies

$$\mathbb{E}^p_t(a_{t+1} - \mu_{t+1}) = \mathbb{E}^m_t a_{t+1} - \mathbb{E}^m_t\big(\mathbb{E}^m_{t+1} a_{t+1}\big) = 0,$$

where I use $\mathbb{E}^p_t = \mathbb{E}^m_t$ under the common information set.

I conjecture that equilibrium inflation and the output gap take the form

$$y_t = \phi_y(a_t - \mu_t), \qquad \pi_t = \phi_\pi(a_t - \mu_t).$$

The conditional-mean result above then implies

$$\mathbb{E}^p_t y_{t+1} = \mathbb{E}^p_t \pi_{t+1} = 0.$$

Substituting the conjectured solution and the optimal policy rule

$$i^*_t = \sigma \mu_t$$

into equation (3) and equation (4) gives

$$\begin{aligned} \phi_\pi(a_t - \mu_t) &= \kappa \phi_y(a_t - \mu_t), \\ \phi_y(a_t - \mu_t) &= -\frac{1}{\sigma} i^*_t + a_t = a_t - \mu_t. \end{aligned}$$

Hence,

$$\phi_y = 1, \qquad \phi_\pi = \kappa,$$

and equilibrium inflation and the output gap are

$$y_t = a_t - \mu_t, \qquad \pi_t = \kappa(a_t - \mu_t) = \kappa y_t.$$

B.5.2. Informational equivalence of the signals under symmetric information

Consider the model under symmetric information. From the equilibrium characterized above,

$$\mathbb{E}_t^p y_{t+1} = \mathbb{E}_t^p \pi_{t+1} = 0,$$

so the equilibrium conditions imply

$$y_t = a_t - \frac{i_t}{\sigma}, \qquad \pi_t = \kappa y_t.$$

Suppose that the monetary authority observes

$$\widetilde{y}_t = y_t + \nu_t^y, \qquad \widetilde{\pi}_t = \pi_t + \nu_t^\pi,$$

where, conditional on the observed uncertainty level ω_t, the measurement errors are mutually independent, independent of the demand shock and past information, and satisfy

$$\nu_t^y, \nu_t^\pi \sim N(0, R_t), \qquad R_t = \chi \omega_t \bar{\sigma}_\varepsilon^2.$$

Because the policy rate associated with the observations is known, I can construct the policy-adjusted signals

$$r_t^y \equiv \widetilde{y}_t + \frac{i_t}{\sigma} = a_t + \nu_t^y, \qquad r_t^\pi \equiv \widetilde{\pi}_t + \frac{\kappa i_t}{\sigma} = \kappa a_t + \nu_t^\pi.$$

I then define the scalar signal

$$z_t \equiv \frac{r_t^y + \kappa r_t^\pi}{1+\kappa^2} = a_t + \eta_t, \qquad \eta_t \equiv \frac{\nu_t^y + \kappa \nu_t^\pi}{1+\kappa^2}.$$

By independence and normality of the measurement errors,

$$\eta_t \mid \omega_t \sim N\left(0, \frac{R_t}{1+\kappa^2}\right) = N\left(0, \frac{\chi}{1+\kappa^2}\omega_t \bar{\sigma}_\varepsilon^2\right).$$

Thus, the scalar signal has the same form as the signal used in the filtering problem above, with

$$\chi_z = \frac{\chi}{1+\kappa^2}.$$

To establish informational equivalence, consider the likelihood of the policy-adjusted

observations conditional on a candidate realization a of the effective demand wedge:

$$L_t(a) \propto \exp\left\{-\frac{(r_t^y - a)^2 + (r_t^\pi - \kappa a)^2}{2R_t}\right\}.$$

Completing the square gives

$$(r_t^y - a)^2 + (r_t^\pi - \kappa a)^2 = (1+\kappa^2)(a - z_t)^2 + \frac{(r_t^\pi - \kappa r_t^y)^2}{1+\kappa^2}.$$

The second term is independent of a and therefore cancels from Bayes' rule. Consequently,

$$L_t(a) \propto \exp\left\{-\frac{(a - z_t)^2}{2R_t/(1+\kappa^2)}\right\},$$

which is precisely the likelihood generated by the scalar signal

$$z_t = a_t + \eta_t, \qquad \eta_t \mid \omega_t \sim N\left(0, \frac{\chi}{1+\kappa^2}\omega_t \bar{\sigma}_\varepsilon^2\right).$$

Therefore, conditional on the known policy rate, observing $\widetilde{y}_t$ and $\widetilde{\pi}_t$ is informationally equivalent to observing the scalar signal z_t. Hence, the monetary authority can solve the filtering problem using z_t alone.

B.5.3. Parameter values and numerical experiment

Table B.1 reports the parameter values and initialization used to construct Figure 1 under the common information set. The demand innovation variance is $Q = \omega\bar{\sigma}_\varepsilon^2$, and each measurement-error variance is $R = \chi Q$. The equivalent scalar signal therefore has noise variance $\chi_z Q$, where $\chi_z = \chi/(1+\kappa^2)$. Uncertainty is held constant within each experiment. The filter starts from the stationary regime prior and retains $J = 100$ initial episode durations, with renormalized weights and zero component means. No subsequent truncation is imposed.

The experiment holds the realized regime at D and sets $a_0 = 1$ in both uncertainty states. Subsequent demand innovations and all realized measurement errors are zero, so $a_t = \rho^t a_0$. Agents do not know these restrictions on the realized path. The figure compares responses to the same absolute shock over $t = 0, \ldots, 15$.

Table B.1: Parameter values and settings for the main-text model figure

Parameter	Description	Value
β	Discount factor	0.99
σ	Inverse intertemporal elasticity of substitution	1
κ	Slope of the New Keynesian Phillips curve	0.024
ρ	Persistence of demand within an active episode	0.8
p_{DD}	Probability of remaining in regime D	0.9
p_{NN}	Probability of remaining in regime N	0.9
p_{DN}	Transition probability from N to D	0.1
p_{ND}	Transition probability from D to N	0.1
$\bar{\sigma}_\varepsilon^2$	Baseline demand innovation variance	1
χ	Each measurement-error variance relative to Q	0.5
ω_L	Low uncertainty	0.25
ω_H	High uncertainty	4
$q_{0\|-1}$	Initial probability of the active regime	0.5
$m_{j,0\|-1}$	Initial mean of each demand component	0
a_0	Initial realized demand shock	1

Notes: The stationary prior is $q_{0|-1} = p_{DN}/(p_{DN} + p_{ND})$. The discount factor does not affect the common-information allocations, since expected future inflation and the output gap are zero.

B.5.4. Private agents have full information

I now assume that private agents observe the current regime s_t and the effective demand wedge a_t, in addition to the monetary authority's information set $\mathcal{I}_t^m$. They can therefore reconstruct the monetary authority's posterior beliefs. The monetary authority observes a noisy signal of the effective demand shock directly, $z_t = a_t + \eta_t$, with the same noise distribution as the equivalent scalar signal derived in Section B.5.2. Its filtering problem is therefore unchanged. Private-sector expectations of future inflation and the output gap, however, are generally nonzero: private agents observe the discrepancy between the true demand wedge and the monetary authority's estimate and account for its expected persistence.

The direct-signal assumption keeps the monetary authority's information independent of its contemporaneous policy choice. Under symmetric information, Section B.5.2 establishes that this signal is informationally equivalent to the observed inflation and output-gap signals. With full information for private agents, inferring the demand shock from inflation and the output gap would instead introduce an additional signaling problem: private-sector expectations would depend on the true state and anticipated pol-

icy, while the monetary authority's signals would depend on equilibrium outcomes and hence on its own policy choice. I abstract from this feedback because it would substantially complicate the non-Gaussian filtering problem and is not directly disciplined by the empirical analysis.

I first establish two useful prediction identities. Recall that

$$\mu_t \equiv \mathbb{E}_t^m a_t, \qquad a_t = d_t \mathbf{1}_{\{s_t = D\}}.$$

The monetary authority's one-step-ahead forecast of the effective demand wedge satisfies

$$\begin{aligned}
\mu_{t+1|t} &\equiv \mathbb{E}_t^m a_{t+1} \\
&= q_t p_{DD} \mathbb{E}_t^m [d_{t+1} \mid s_t = D, s_{t+1} = D] \\
&\quad + (1 - q_t) p_{DN} \mathbb{E}_t^m [d_{t+1} \mid s_t = N, s_{t+1} = D] \\
&= \rho p_{DD} q_t m_t = \rho p_{DD} \mu_t.
\end{aligned}$$

The $N \to D$ term is zero because a new mean-zero demand shock is drawn upon entry into regime D, while transitions into regime N contribute zero because the effective demand wedge vanishes in that regime.

Private agents instead observe the current regime and the effective demand wedge. Therefore,

$$\mathbb{E}_t^p a_{t+1} = \begin{cases} \rho p_{DD} a_t, & s_t = D, \\ 0, & s_t = N. \end{cases}$$

Since $a_t = 0$ in regime N, this expression can be written compactly as

$$\mathbb{E}_t^p a_{t+1} = \rho p_{DD} a_t.$$

Thus, private-sector and monetary-authority forecasts differ whenever $a_t \neq \mu_t$.

Unlike under common information, the law of iterated expectations does not imply

$$\mathbb{E}_t^p (a_{t+1} - \mu_{t+1}) = 0.$$

The reason is that

$$\mu_{t+1} = \mathbb{E}_{t+1}^m a_{t+1}$$

is defined using the monetary authority's information set, which does not contain the private sector's information at time t. Consequently, the equilibrium allocation derived in Section B.5.1 does not generally apply.

To characterize the recursive equilibrium, I define the monetary authority's posterior belief as

$$b_t \equiv \left(1 - q_t, \left\{w_{j,t|t}, m_{j,t|t}, P_{j,t|t}\right\}_{j\geq 1}\right).$$

Equivalently, the belief regime can be represented by $(q_t, f_{t|t})$. Aggregate moments alone are not sufficient to determine the next filtering update. I therefore define the state

$$x_t = (s_t, a_t, b_t)$$

and write equilibrium output and inflation as

$$y_t = Y(x_t), \qquad \pi_t = \Pi(x_t).$$

Let $\mathcal{F}(b_t, z_{t+1})$ denote the monetary authority's prediction and updating rule, so that

$$b_{t+1} = \mathcal{F}(b_t, z_{t+1}).$$

Conditional on the current state, s_{t+1} follows the Markov transition probabilities, and the effective demand wedge evolves according to

$$a_{t+1} = \begin{cases} \rho a_t + \varepsilon_{t+1}, & s_t = D,\ s_{t+1} = D, \\ \varepsilon_{t+1}, & s_t = N,\ s_{t+1} = D, \\ 0, & s_{t+1} = N, \end{cases}$$

where

$$\varepsilon_{t+1} \sim N\left(0, \omega\bar{\sigma}_\varepsilon^2\right).$$

The next-period signal is

$$z_{t+1} = a_{t+1} + \eta_{t+1}, \qquad \eta_{t+1} \sim N\left(0, \chi_z\omega\bar{\sigma}_\varepsilon^2\right),$$

with η_{t+1} independent of ε_{t+1} and the regime transition.

These laws of motion define the private-sector expectation operator

$$(\mathcal{T}G)(x_t) \equiv \mathbb{E}_t^p[G(s_{t+1}, a_{t+1}, \mathcal{F}(b_t, z_{t+1}))]$$

for any function G. Because the state transition and signal process do not depend on the allocation functions or the policy rate, the filtering problem can be solved separately from the allocation problem.

Given the optimal policy rule

$$i_t^* = \sigma \mu_t, \qquad \mu_t = \sum_j w_{j,t|t} m_{j,t|t} \equiv \mu(b_t),$$

the equilibrium conditions can be written recursively as

$$\begin{aligned} Y(x) &= a - \mu(b) + (\mathcal{T}Y)(x) + \frac{1}{\sigma}(\mathcal{T}\Pi)(x), \\ \Pi(x) &= \kappa Y(x) + \beta(\mathcal{T}\Pi)(x). \end{aligned}$$

Thus, although the allocation functions are generally nonlinear in the state, the equilibrium equations are linear in the unknown functions Y and Π for a given transition operator $\mathcal{T}$.

Computational implementation. For a fixed uncertainty level ω and the initialization described above, the component variances do not depend on the realized signals. In particular,

$$P_{j,t|t} = \frac{\chi_z \omega \bar{\sigma}_\varepsilon^2 P_{j,t|t-1}}{P_{j,t|t-1} + \chi_z \omega \bar{\sigma}_\varepsilon^2}, \qquad P_{j+1,t+1|t} = \rho^2 P_{j,t|t} + \omega \bar{\sigma}_\varepsilon^2.$$

Therefore, the component variances and Kalman gains can be precomputed and need not be included as continuous coordinates of the numerical state. They are determined by episode duration and, for episodes already active at initialization, by their initial duration, whereas component means and posterior weights remain history dependent.

To solve the recursive equilibrium numerically, I approximate the belief state by a finite representation and the expectation operator $\mathcal{T}$ by quadrature and interpolation. Because the posterior mixture weights depend on the future signal, the expectation operator integrates jointly over the next-period regime, demand innovation, and signal noise while updating the monetary authority's beliefs at each possible realization.

Let $T_{i\ell}$ denote the resulting quadrature and interpolation weights from state node x_i to node x_ℓ. The discretized equilibrium conditions are

$$\begin{aligned} Y_i &= \sum_\ell T_{i\ell} Y_\ell - \mu_i + \frac{1}{\sigma} \sum_\ell T_{i\ell} \Pi_\ell + a_i, \\ \Pi_i &= \kappa Y_i + \beta \sum_\ell T_{i\ell} \Pi_\ell. \end{aligned}$$

In matrix notation,

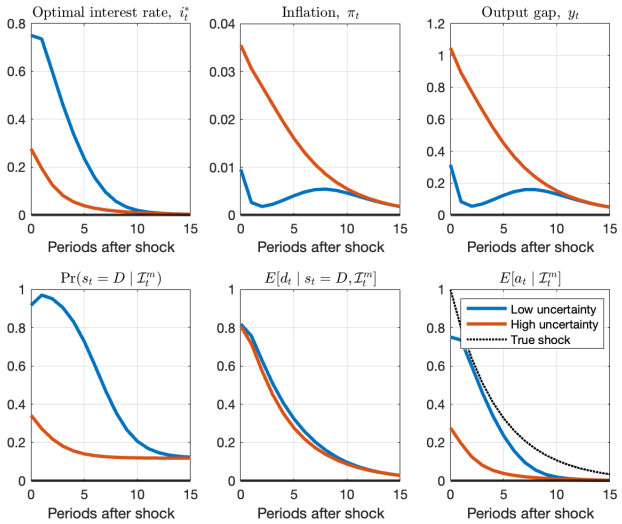
$$\begin{bmatrix} I - T & -T/\sigma \\ -\kappa I & I - \beta T \end{bmatrix} \begin{bmatrix} Y \\ \Pi \end{bmatrix} = \begin{bmatrix} a - \mu \\ 0 \end{bmatrix}.$$

Here, T approximates expectations over the full belief state rather than an exact transition matrix for a lower-dimensional state vector. The accuracy of the numerical solution therefore depends on the finite representation of the belief state, the state discretization, and the numerical integration used to construct T.

B.5.5. Impulse response functions under full information of private agents

Figure B.1 shows the impulse responses to a positive demand shock when regime D remains active throughout the exercise. The monetary authority does not observe the regime, whereas the private sector has full information. The figure compares the responses under two levels of uncertainty, ω, which is held fixed over time and known to all agents. The blue line corresponds to low uncertainty, while the red line corresponds to high uncertainty.

Figure B.1: Impulse responses to a persistent demand shock under low and high uncertainty with full information for private agents



Notes: The blue line denotes low uncertainty and the red line high uncertainty. The top panels report the responses of the policy rate, inflation, and the output gap. The bottom panels report the monetary authority's posterior probability of the active demand regime, $\Pr(s_t = D \mid \mathcal{I}_t^m)$, the perceived demand shock conditional on the active regime, $\mathbb{E}[d_t \mid s_t = D, \mathcal{I}_t^m]$, and the estimated effective demand shock, $\mathbb{E}[a_t \mid \mathcal{I}_t^m]$. Higher uncertainty lowers the posterior probability of the active demand regime and the estimated effective demand shock, resulting in a weaker policy response and larger responses of inflation and the output gap.

The top left panel shows the impulse response of the interest rate. The monetary authority raises the policy rate by less when uncertainty is higher, as the red line remains below the blue line throughout the response horizon. The top middle panel reports the response of inflation, which is larger under high uncertainty. The top right panel shows the response of the output gap, which is also larger under high uncertainty. Consistent with Figure 1, these panels show that higher uncertainty weakens the monetary authority's response to the demand shock, allowing the shock to propagate more strongly through the economy and generating larger responses of inflation and the output gap.

The bottom left panel of Figure B.1 reports the monetary authority's posterior probability that the demand regime is active, $\Pr(s_t = D \mid \mathcal{I}_t^m)$. This probability is lower under high uncertainty. The bottom middle panel reports the perceived demand shock conditional on the active regime, $\mathbb{E}[d_t \mid s_t = D, \mathcal{I}_t^m]$, which is very similar across the two levels of uncertainty. This similarity reflects the fact that uncertainty scales the variances of both the demand innovation and the measurement error proportionally, leaving their relative precision unchanged. Finally, the bottom right panel reports the monetary authority's estimate of the effective demand shock, $\mathbb{E}[a_t \mid \mathcal{I}_t^m]$. Under higher uncertainty, this estimate is smaller because the monetary authority assigns a lower probability to the active demand regime. The resulting weaker policy response allows the same realized demand shock to generate larger responses of inflation and the output gap.

The bottom panels of Figure B.1 are identical to those of Figure 1. This follows from the assumption that the signal observed by the monetary authority is independent of equilibrium outcomes, so the filtering problem is identical under the two information structures. As a result, the monetary authority forms the same beliefs and chooses the same interest-rate path in the two figures. Inflation and the output gap, however, respond more strongly in Figure B.1 at each level of uncertainty. When private agents have full information, they observe the true persistent demand disturbance and therefore form positive expectations of future inflation and the output gap. These expectations amplify the equilibrium responses of both variables relative to the common-information case. Importantly, the effect of uncertainty remains unchanged: higher uncertainty weakens the policy response and leads to larger responses of inflation and the output gap.

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